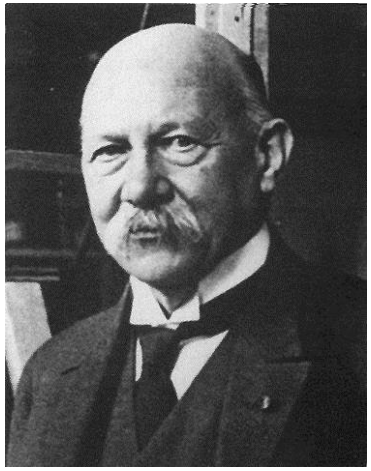
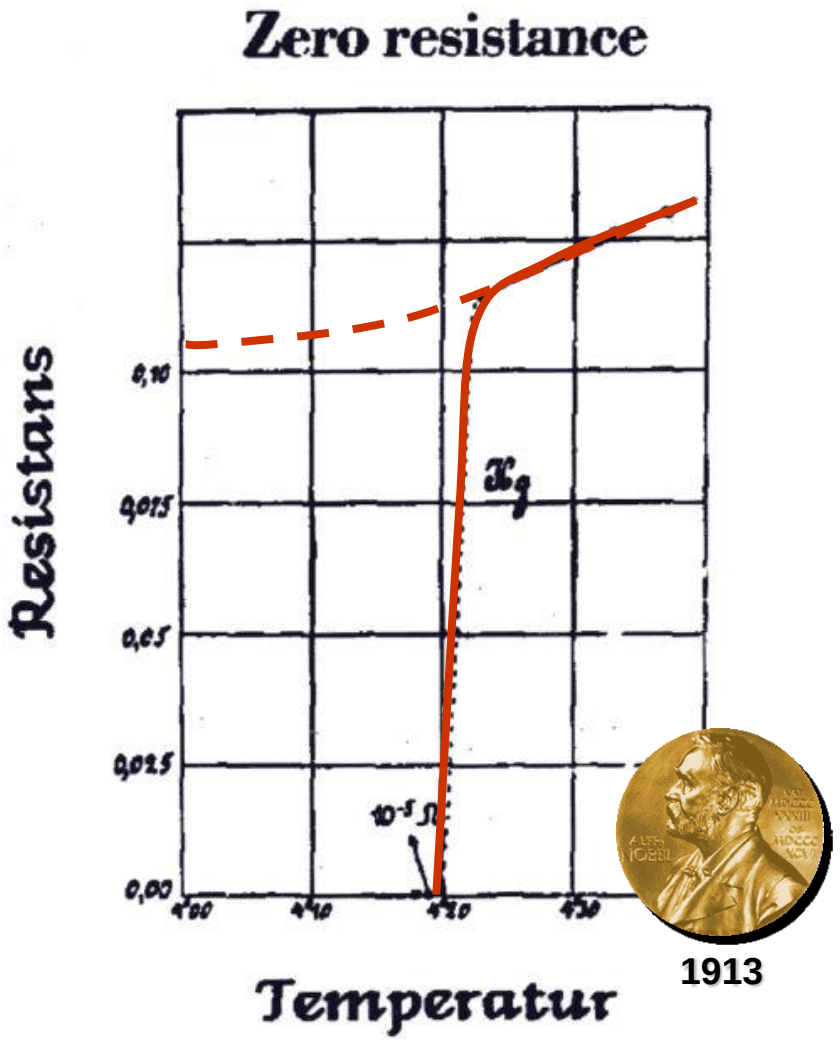
Superconductivity

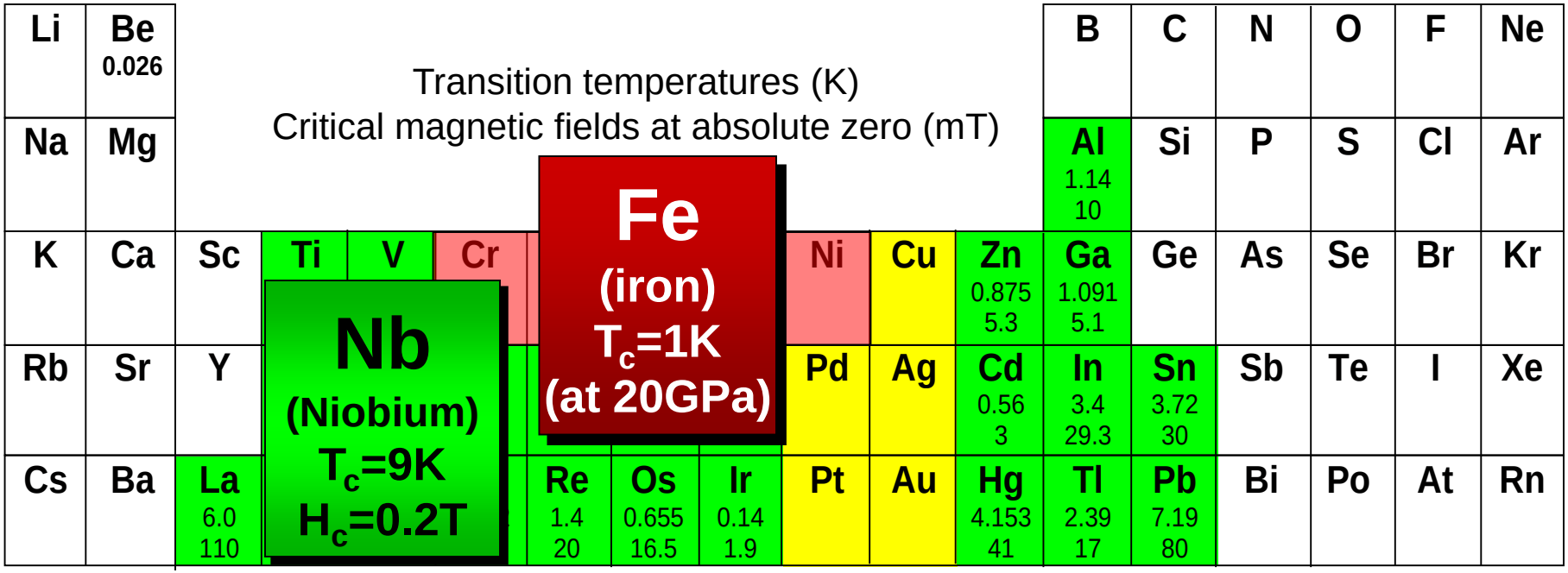
Discovery of Superconductivity



- Discovered by **Kamerlingh Onnes** in 1911 during first low temperature measurements to liquefy helium
- Whilst measuring the resistivity of “pure” Hg he noticed that the electrical resistance dropped to zero at 4.2K
- In 1912 he found that the resistive state is restored in a magnetic field or at high transport currents



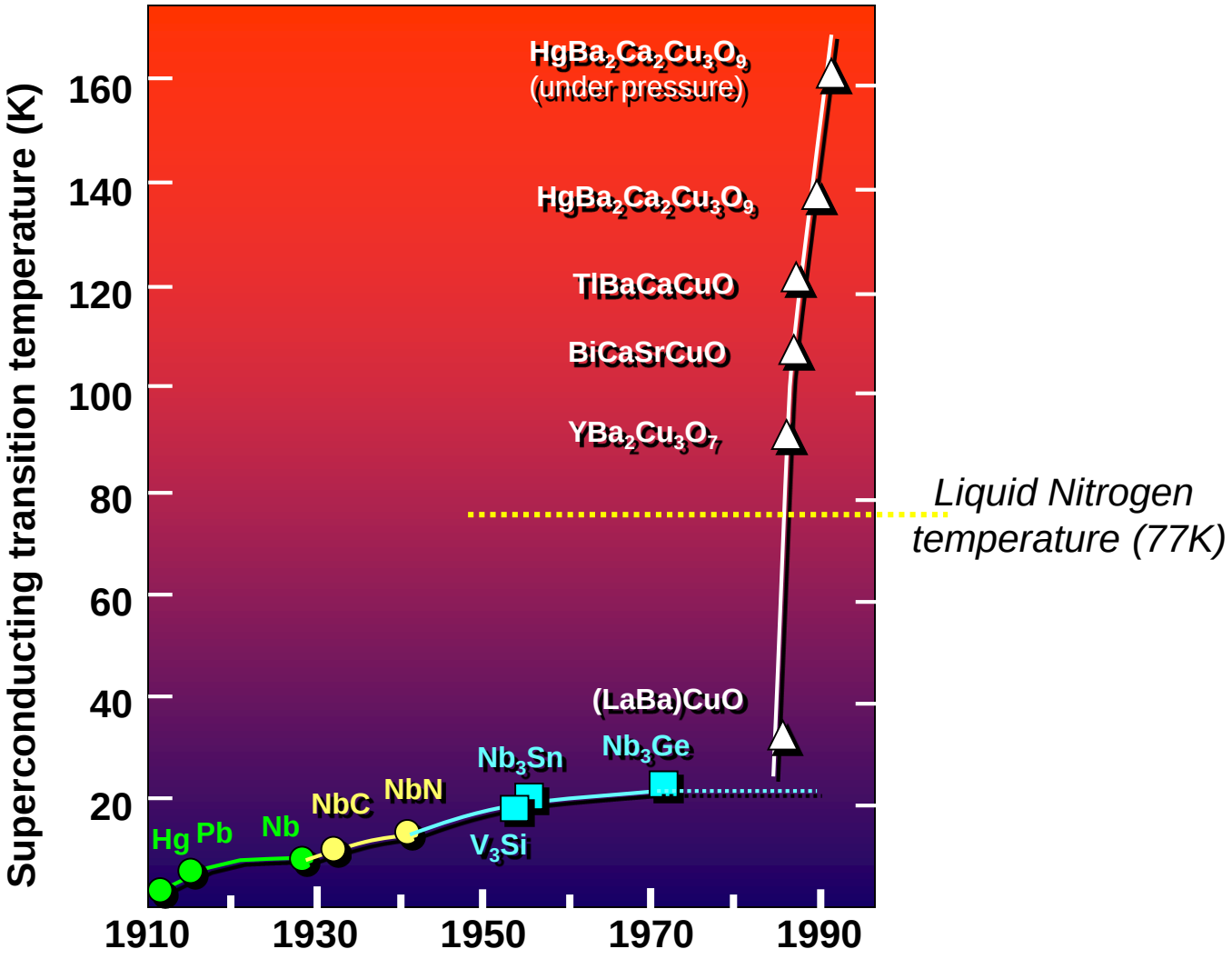
The superconducting elements



- Transition temperatures (K) and critical fields are generally low
- Metals with the highest conductivities are not superconductors
- The magnetic 3d elements are not superconducting

...or so we thought until 2001

Superconductivity in alloys and oxides



Zero resistance?

In a metal a current is carried by free conduction electrons - ie by plane waves

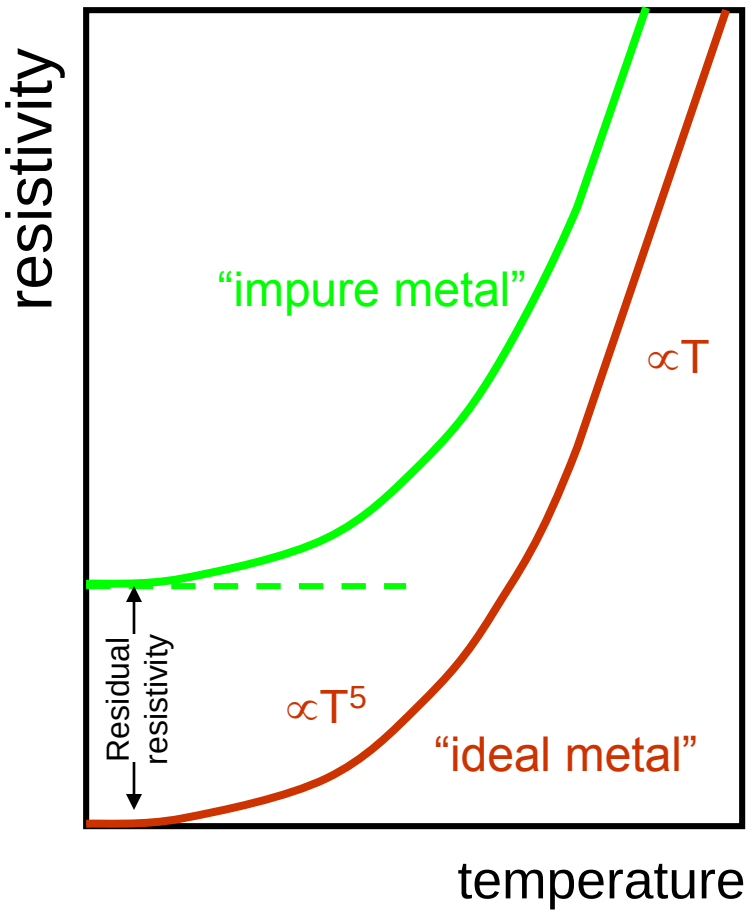
Plane waves can travel through a perfectly periodic structure without scattering…..

….but at finite temperatures phonons destroy the periodicity and cause resistance

Even at T=0, defects such as grain boundaries, vacancies, even surfaces give rise to residual resistivity

Take, eg, pure copper with a resistivity at room temperature of 2μΩcm, and a residual resistivity at 4.2K of 2×10^{-5} μΩcm

…….a typical Cu sample would thus have a resistance of only 2×10^{-11} Ω at 4.2K



Zero resistance?

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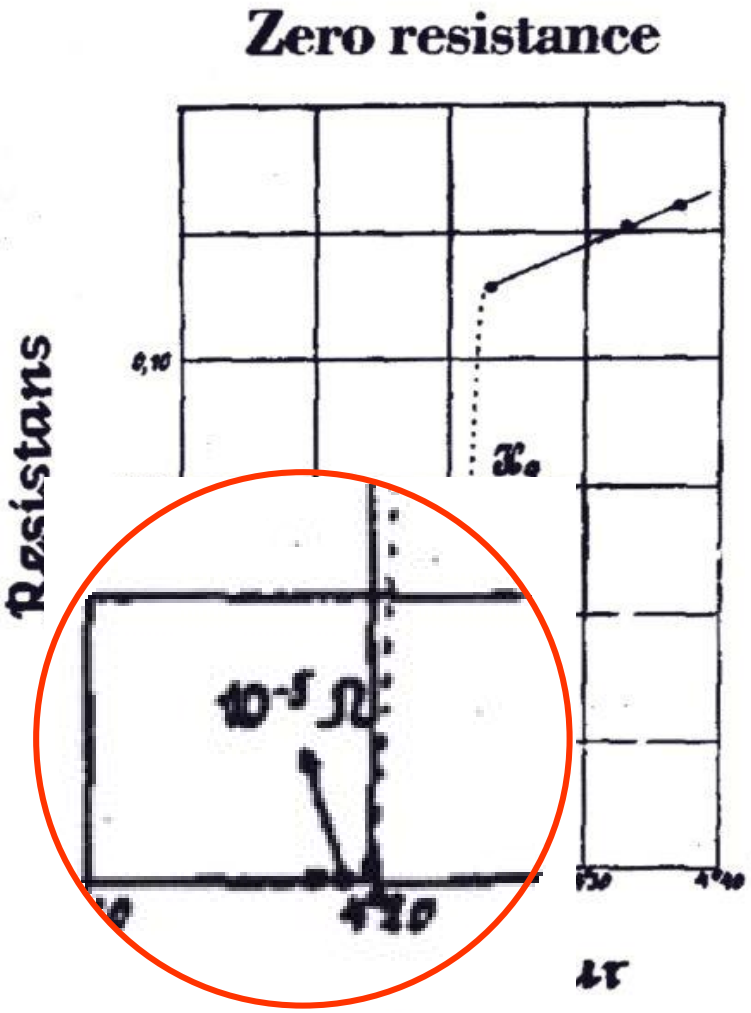
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Zero resistance
Resistans
$10^{-5}\,\Omega$
4,20
T_c
4,20
4,20
4,20

Zero resistance?

The resistance of pure copper is so small is there really much difference between it and that of a superconductor?

Take an electromagnet consisting of a 20cm diameter coil with 10000 turns of 0.3mmx0.3mm pure copper wire

$R_{300K} = 1\ k\Omega$ $R_{4.2K} = 0.01\ \Omega$

Pass a typical current of 20 Amps through the coil

$P_{300K} = 0.4MW$ $P_{4.2K} = 4$ Watts

At 4.2K this is more than enough to boil off the liquid helium coolant!

Measuring zero resistance

Can we determine an upper limit for the resistivity of a superconductor?

This is done by injecting current into a loop of superconductor

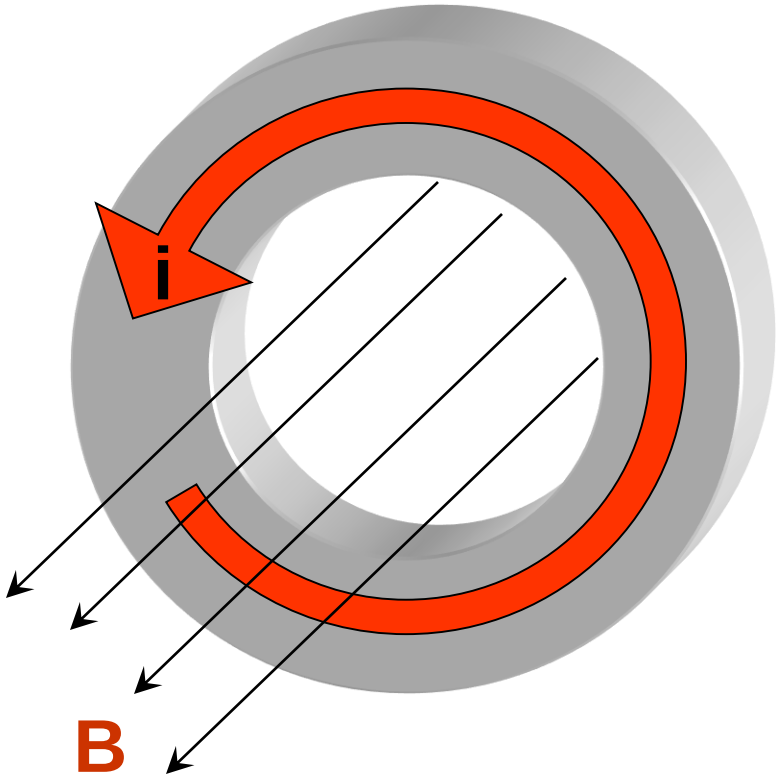
The current generates a magnetic field, and the magnitude of this field is measured as a function of time

This enables the decay constant of the effective R-L circuit to be measured:

$$B(t) \propto i(t) = i(0)e^{-(R/L)t}$$

Using this technique, no discernable change in current was observed over two years:

$$\rho_{sc} \leq 10^{-24} \mu\Omega.\text{cm} \quad !!$$



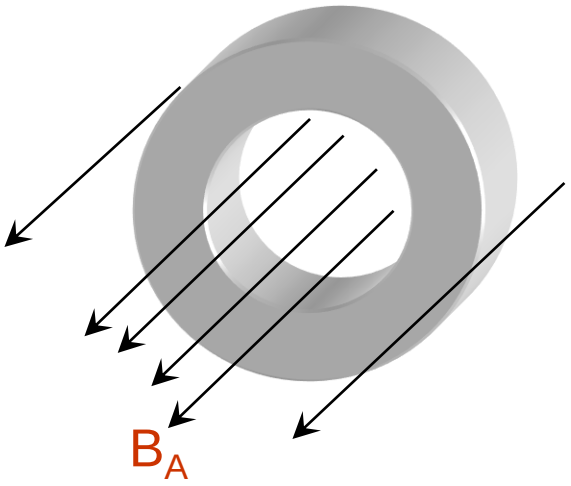
Measuring zero resistance

In practice the superconducting ring is cooled in a uniform magnetic field of flux density B_A to below T_C

If the area of the ring is A, the flux threading the loop is

$$\phi = AB_A$$

Now change B_A: by Lenz's law a current will flow to oppose the change, hence



Cool the ring in an applied magnetic field - then decrease the field to zero

Measuring zero resistance

In practice the superconducting ring is cooled in a uniform magnetic field of flux density B_A to below T_C

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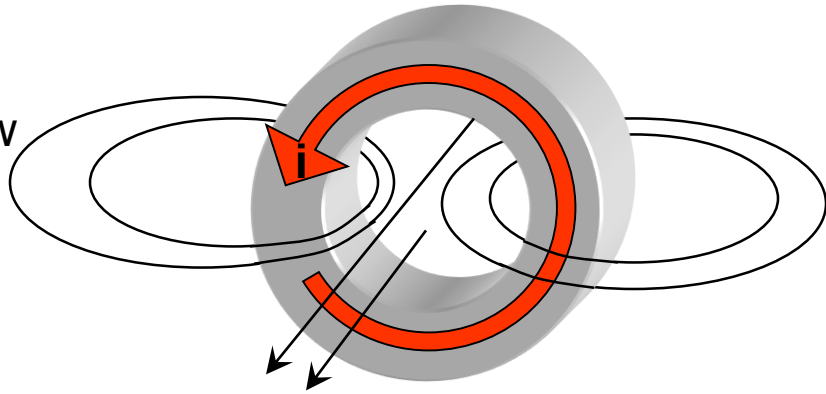
$$\text{emf} \quad -A\frac{dB_A}{dt} = Ri + L\frac{di}{dt}$$

In a "normal" loop, the Ri term quickly kills the current, but if R=0

$$-A\frac{dB_A}{dt} = L\frac{di}{dt}$$

Therefore $Li+AB_A$ = constant (=total flux in loop)

So if R=0 the current will persist forever !!



Currents will flow to maintain the field in the loop…. forever

If $-A\frac{dB_A}{dt} = Ri + L\frac{di}{dt}$

and $Ri = 0$ such that

$Li+AB_A$ = constant (=total flux in loop)

The flux in the superconducting loop must remain constant however the field changes

Therefore if a loop is cooled into the superconducting state in zero field and *then the magnetic field is applied* supercurrents must circulate to maintain the total flux threading the loop at zero.

A superconducting cylinder can therefore provide perfect magnetic shielding

A Meissner Shield

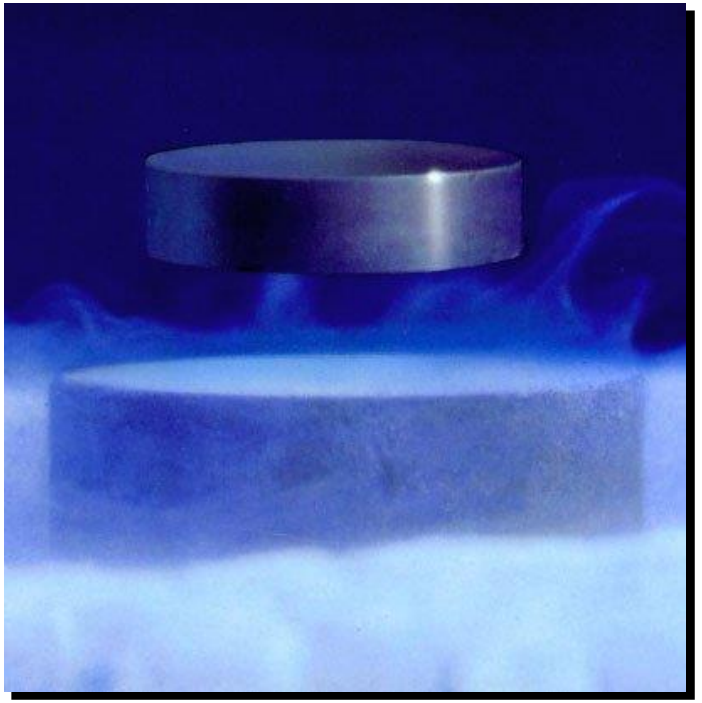
The Meissner Effect

So far everything we have discussed is equally true for a "perfect conductor" as well as a "superconductor"

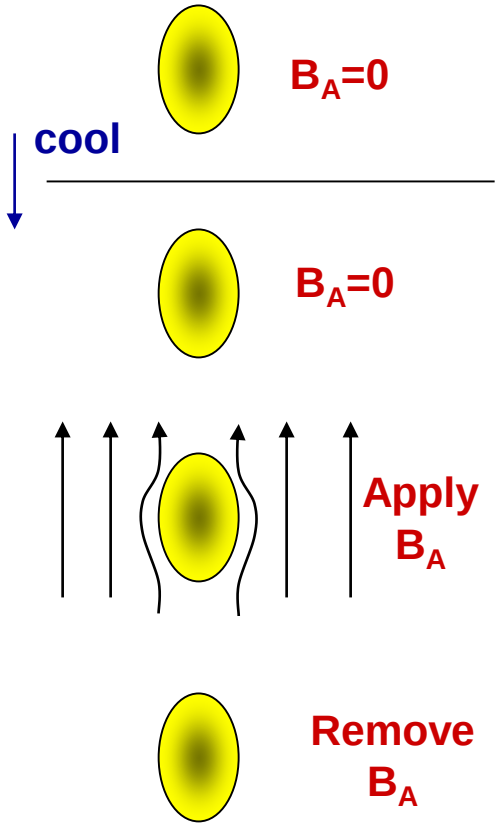
In 1933 Meissner and Oschenfeld made a discovery which distinguished between the two

The Meissner Effect

"A superconductor excludes all magnetic flux from its interior"



A “perfect conductor” - cooled in zero field



The perfect conductor is cooled in zero magnetic flux density to below “T_c”

dB/dt must be zero in a closed resistanceless loop so screening currents flow to generate a field equal and opposite to B_A within the perfect conductor

As B_A is reduced to zero, dB/dt must remain at zero, so the screening currents also decrease to zero.

A “perfect conductor” - cooled in a field

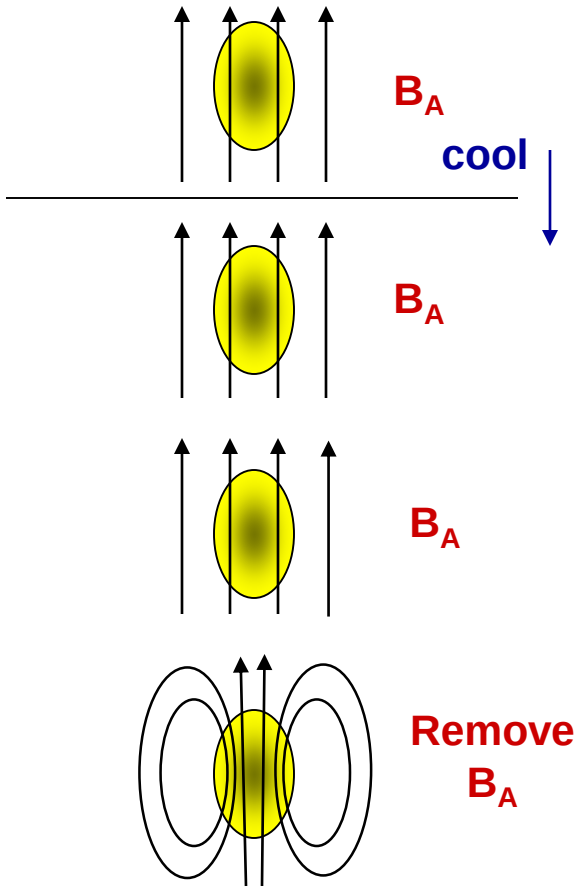
A magnetic flux density B_A is applied to the perfect conductor at high temperatures

It is then cooled in a magnetic flux density B_A to below “T_c”

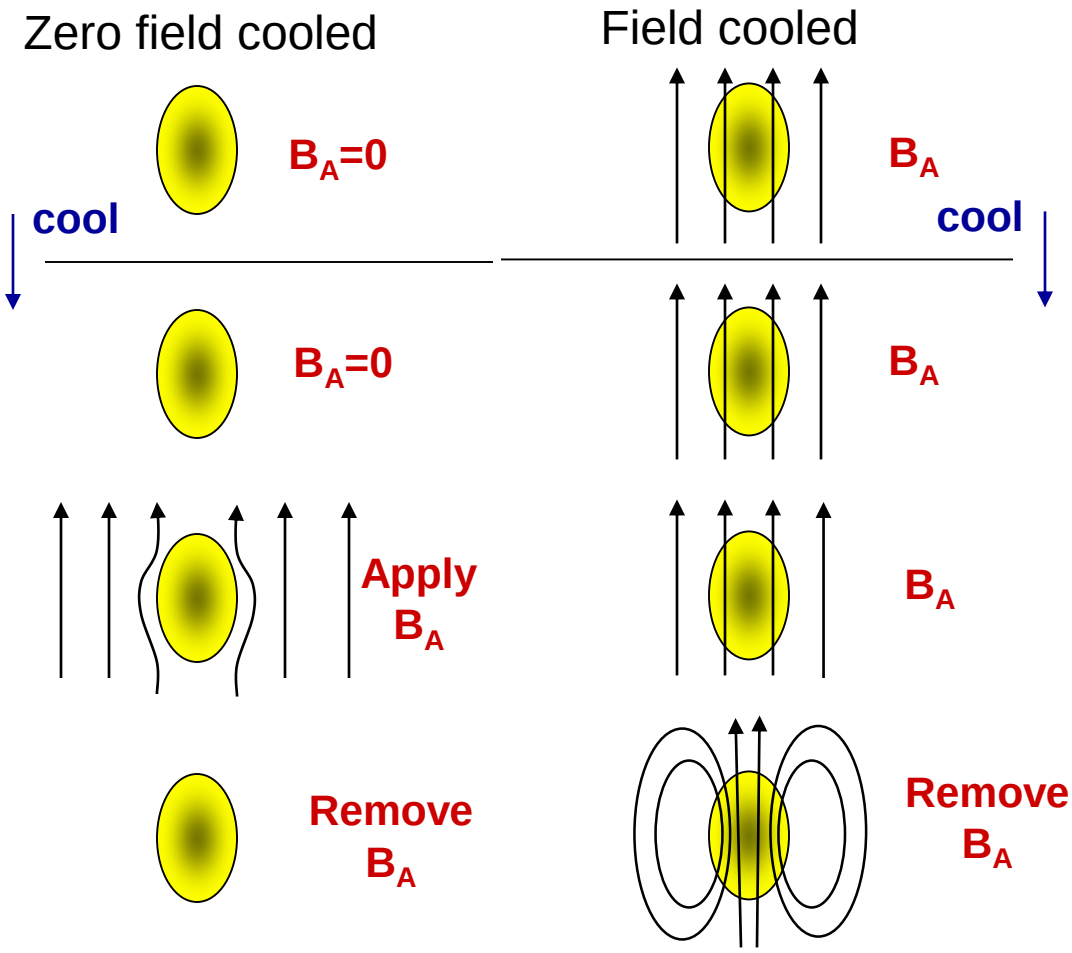
Because there is no change in flux density within the perfect conductor dB/dt=0 and no screening currents flow. B_A is maintained within the sample

As B_A is reduced to zero, screening currents flow. In order to ensure dB/dt=0 and hence to maintain a flux density of B_A within the sample

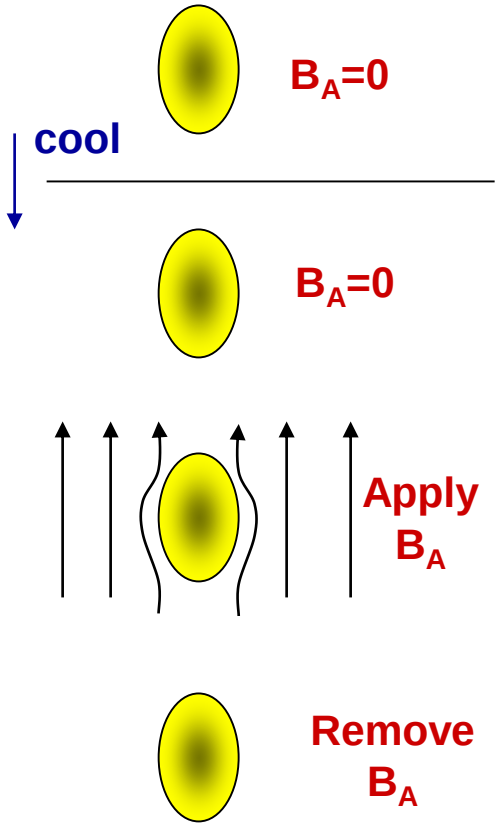
The currents continue to flow even when the applied flux density is reduced to zero - the sample is effectively magnetised



A “perfect conductor”



A superconductor - cooled in zero field



The superconductor is cooled in zero magnetic flux density to below "T_c"

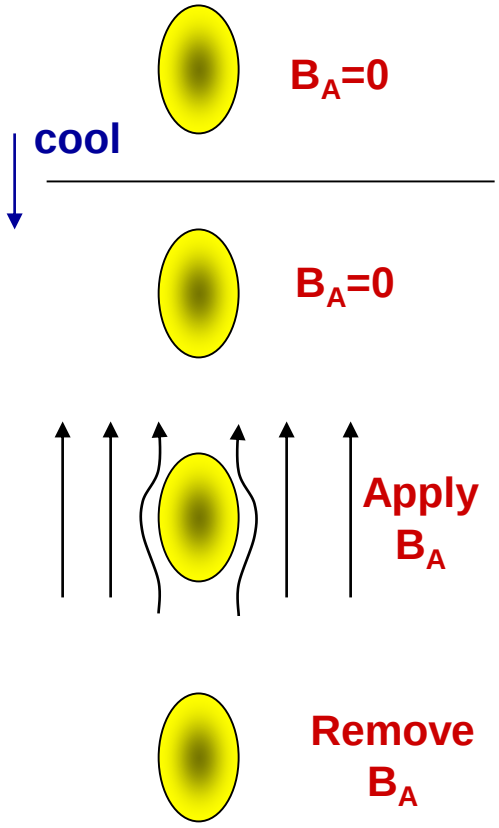
dB/dt must be zero in a closed resistanceless loop so screening currents flow to generate a field equal and opposite to B_A within the superconductor

As B_A is reduced to zero, dB/dt must remain at zero, so the screening currents also decrease to zero.

Precisely the same as a perfect conductor

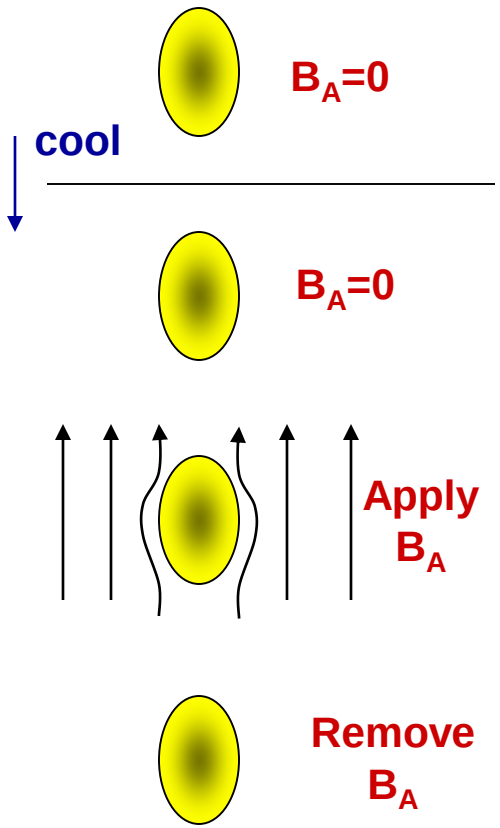
superconductor

Zero field cooled



perfect conductor

Zero field cooled



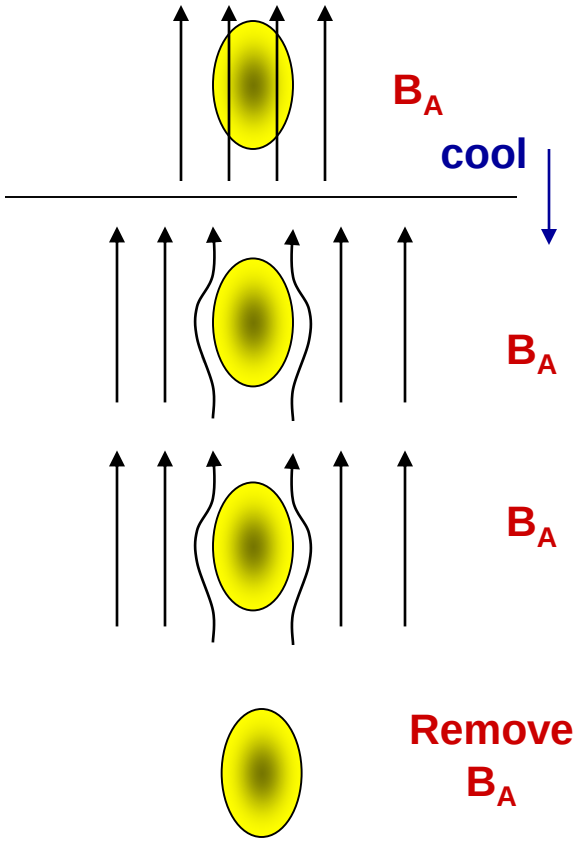
A superconductor” - cooled in a field

A magnetic flux density B_A is applied to the superconductor at high temperatures

It is then cooled in a magnetic flux density B_A to below “T_c”

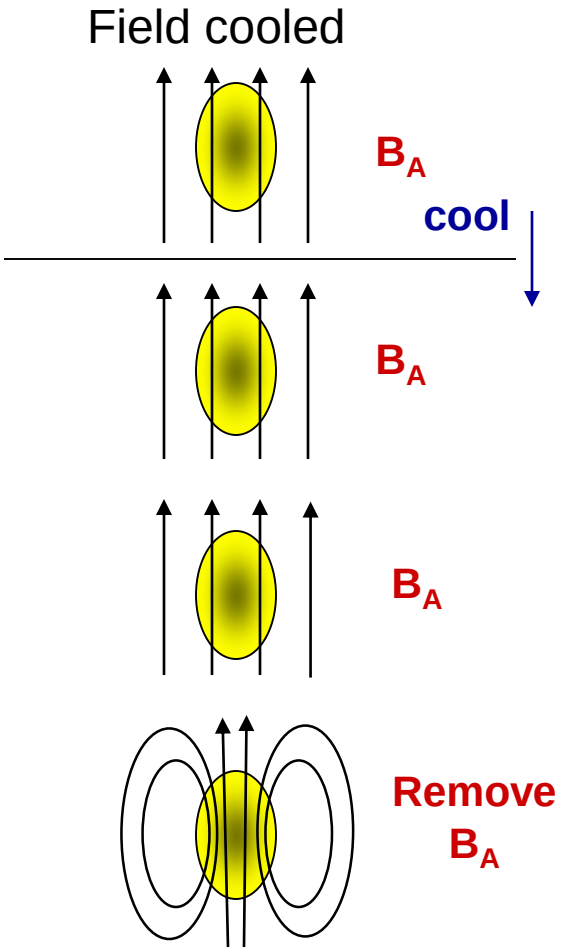
All magnetic flux is spontaneously excluded from the body of the superconductor - even though the applied flux density is unchanged and dB/dt=0 . Screening currents must therefore begin flow in a time invariant field to produce fields equal and opposite to B_A!!

As the applied magnetic flux density is reduced to zero, the screening currents also decrease to ensure that dB/dt=0 within the superconductor.

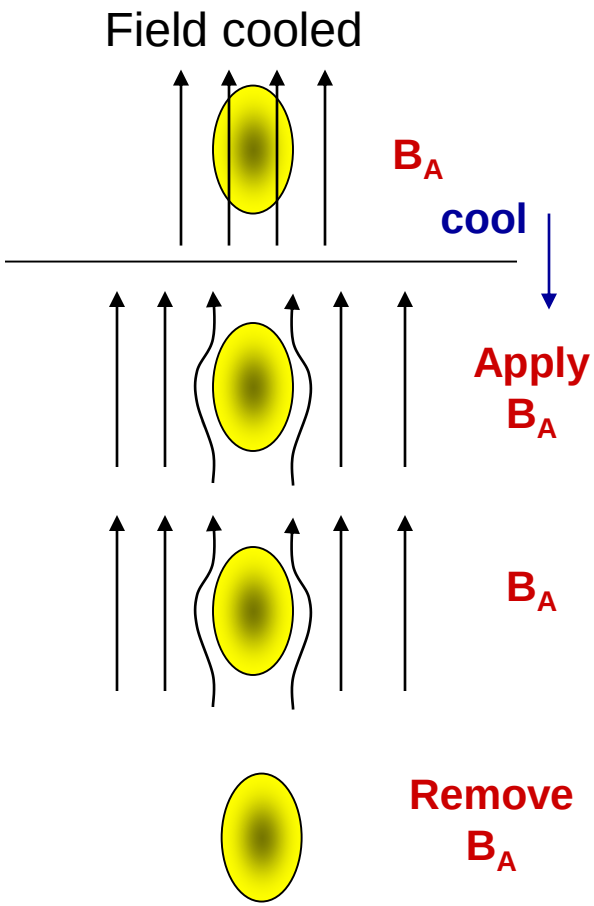


This is the Meissner Effect - it shows that not only must dB/dt=0 within a superconductor - but B itself must remain zero

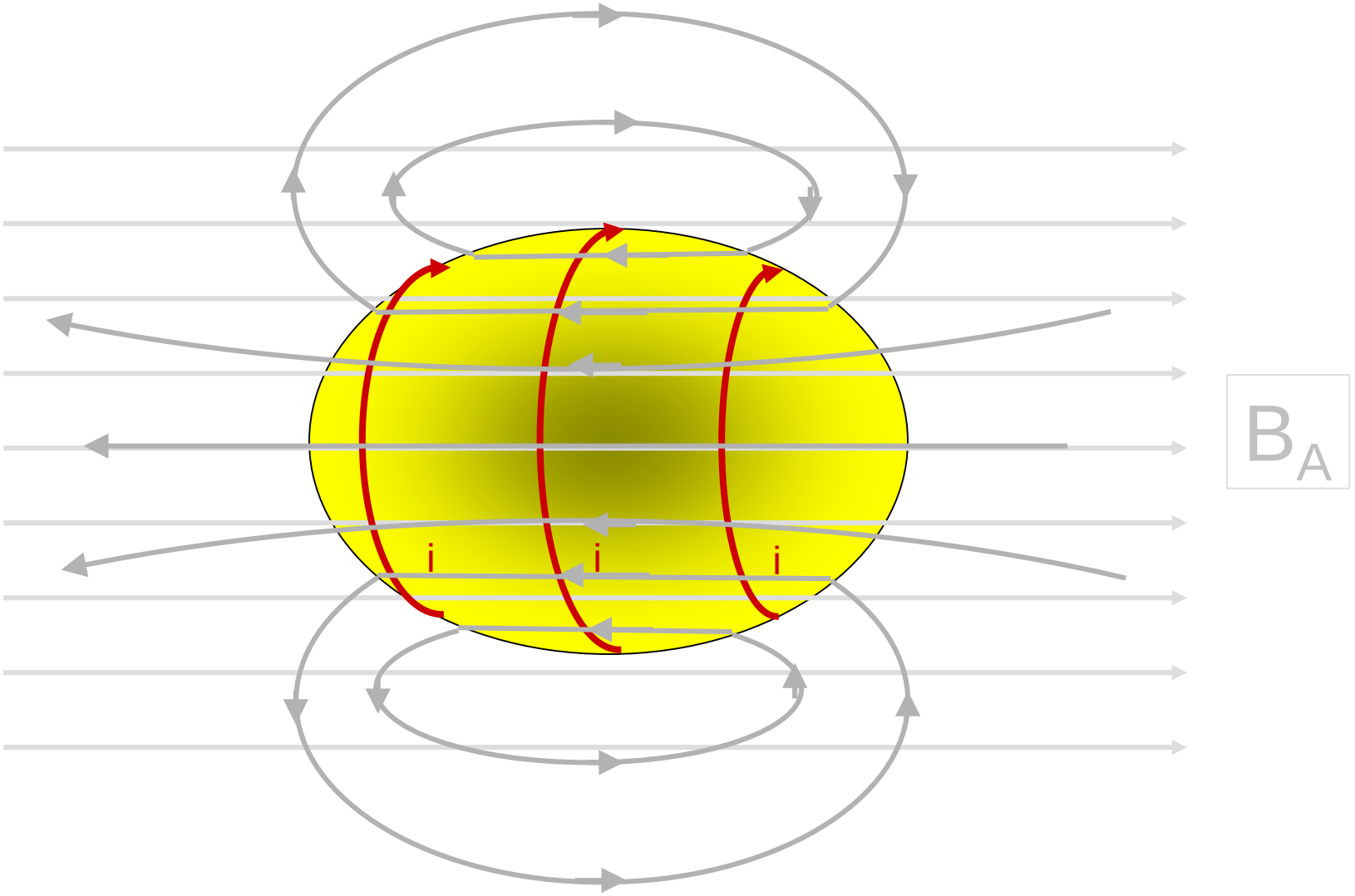
perfect conductor



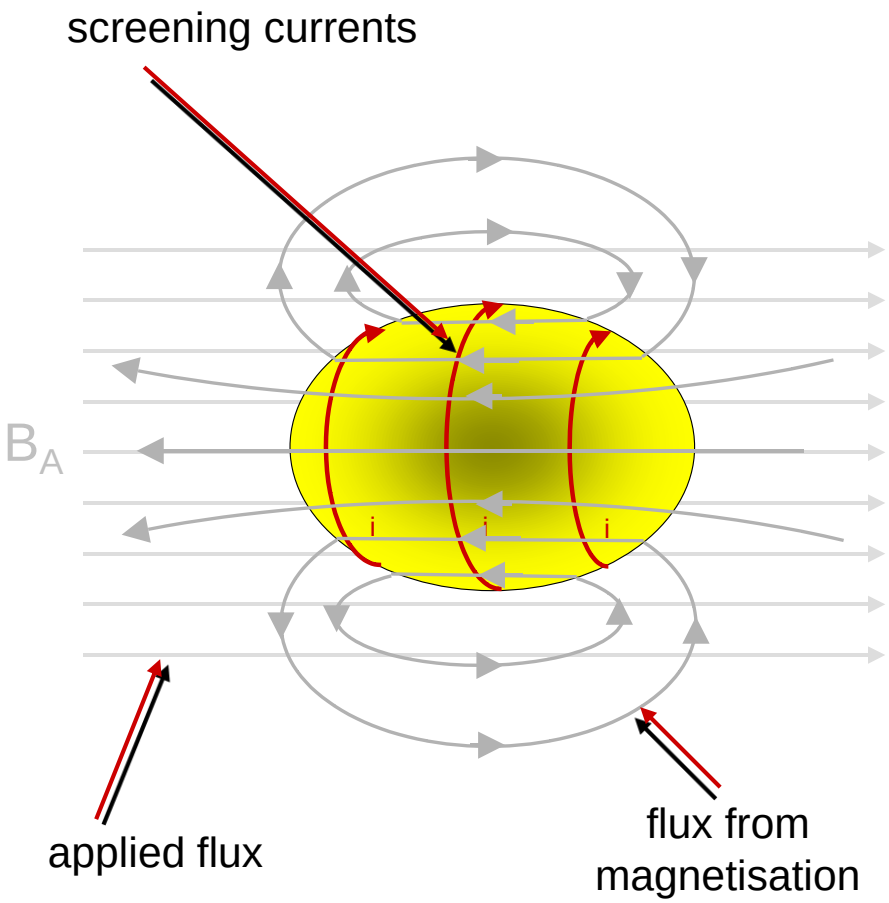
superconductor

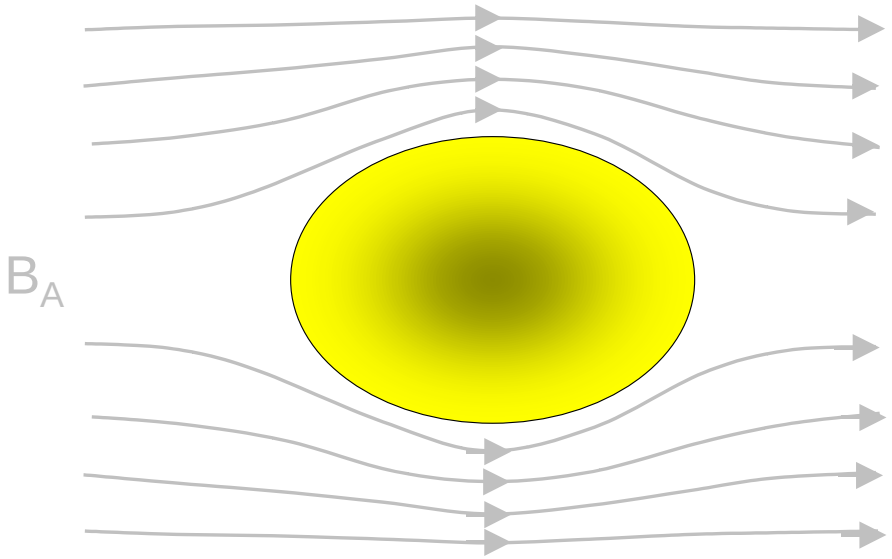


Screening currents - solid sample



Net flux distribution - solid sample





An example of perfect diamagnetism

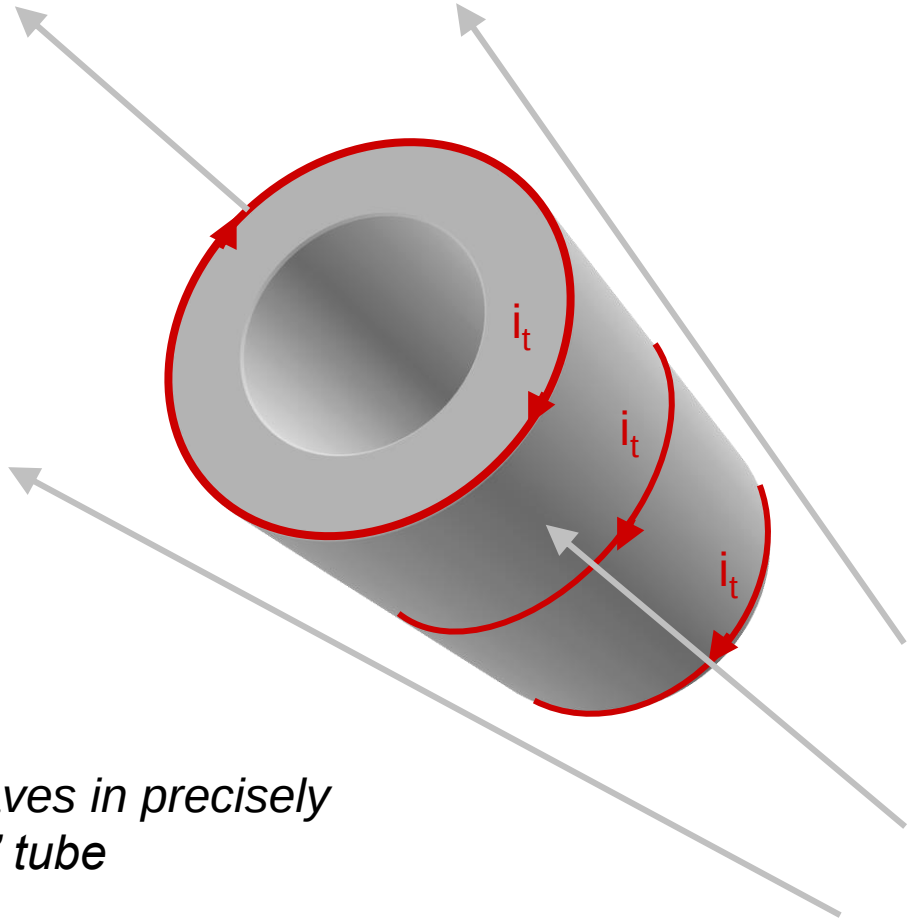
A tube - (a simply connected system)

Magnetic field applied after cooling superconducting tube in zero field:

B=0 within the body of the material

On application of field, B is maintained at zero by circulation of screening currents i_t on outer surface

i_t also cancels the flux density due to applied field in the hole



In this case a superconducting tube behaves in precisely the same way as a "perfectly conducting" tube

A tube - (a simply connected system)

Cooling a superconducting tube in an applied magnetic field:

Above T_C the flux passes through the body of the tube and the hole

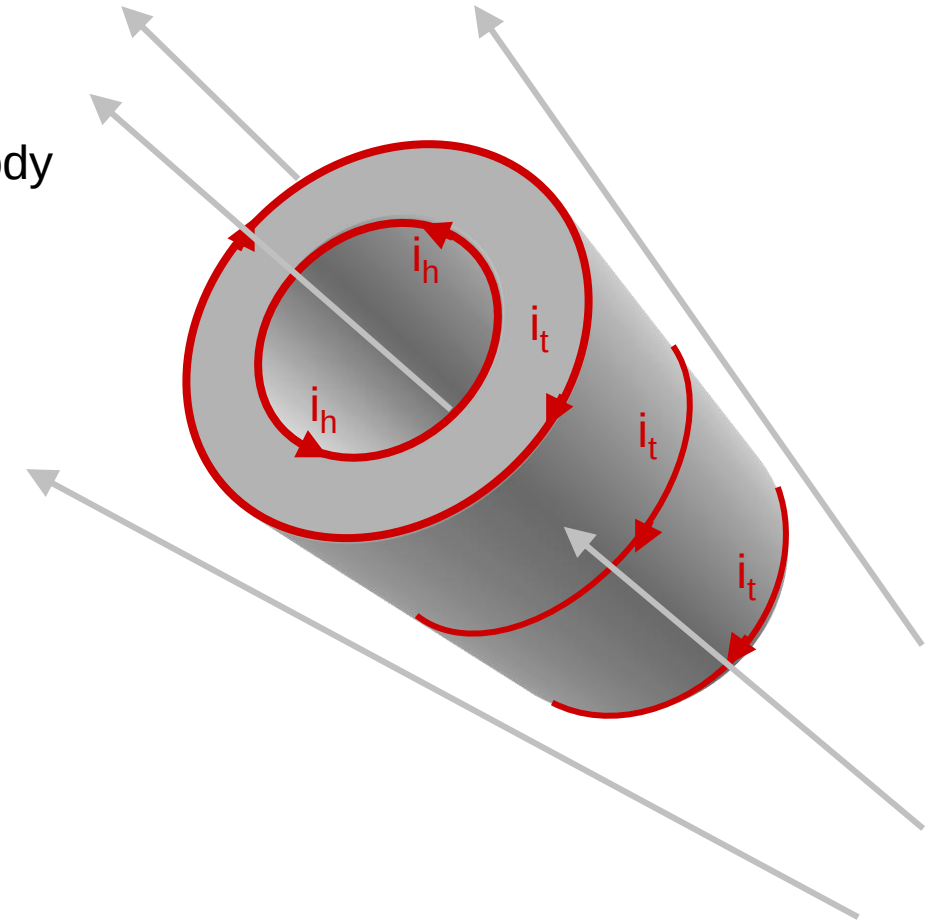
On cooling into the superconducting state, flux is expelled from body of tube

Circulation of screening currents i_t on outer surface ensures B=0 in the body of the tube

However i_t also cancels the flux density due to applied field in the hole….

….but the hole is not a superconductor
- the flux density must not change!

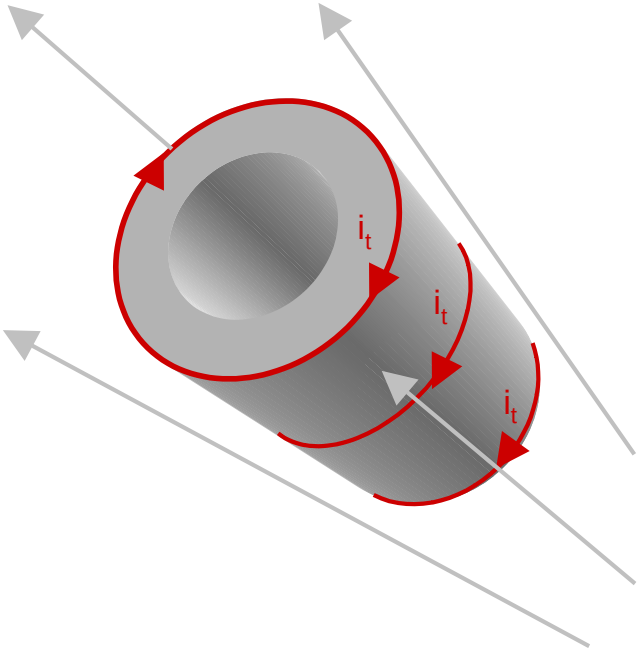
Therefore currents i_h must flow at the *inner surface* of the tube to preserve the flux density in the hole



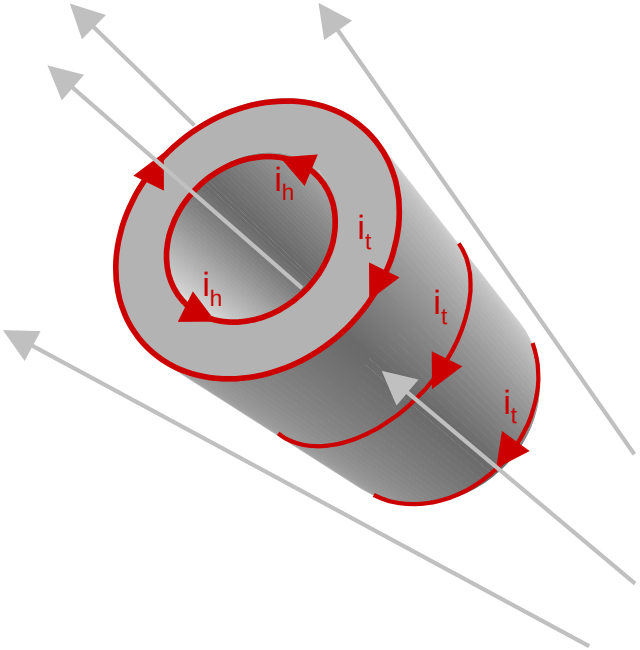
Question: how would a perfectly conducting tube behave?

Summary:

Magnetic field applied after cooling superconducting tube in zero field:



Cooling a superconducting tube in an applied magnetic field:



Note that $\mathbf{i_t - i_h}$ maintains a value which generates a flux density just equal to the difference between the flux density in the hole and outside the superconducting body

Even if the applied field is now reduced to zero, the field within the tube (which is now generated by $\mathbf{i_h}$) will persist

The Meissner Effect - summary

Between 1911 and 1933 researchers considered that a superconductor was no more than a resistanceless perfect conductor

By measuring the properties of a superconductor cooled in a magnetic field they showed that not only

$$dB/dt=0 \text{ but also } B=0.$$

The ability of a superconductor to expel magnetic flux from its interior is the Meissner Effect

It is the first indication that the superconducting state is an entirely new state of matter

It shows that in a superconductor currents can be induced to flow in a time invariant field - in violation of Maxwell's equations

Summary: *Superconductors expel all magnetic flux and exhibit zero resistance*

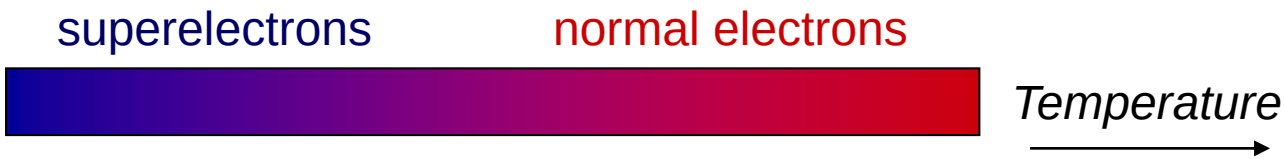
Direct currents

Zero resistance implies no voltage drop across a superconductor

Therefore irrespective of length no power is generated!

This is really true only if the current is dc

A superconductor can be considered as a mixture of two “fluids”



At T=0 all electrons are superelectrons, for $T>T_c$ all electrons are normal, with superelectrons converting to normal electrons as T_c is approached

The dc current must be carried by the superelectrons, and there must be no electric field, otherwise the superelectrons would continue to accelerate and the current would increase

The normal electrons are effectively “shorted out”

Alternating currents

If an ac voltage is applied across a superconductor there will be a time varying electric field

Superelectrons, like normal electrons, have mass and hence inertia

So, the supercurrent lags the electric field and therefore produces an inductive impedence

An inductive impedence in turn implies that there is an electric field present, so the normal electrons also carry some current

The superconductor is therefore resistive, and appears as a perfect inductance in parallel with a resistance

The inductive component is small (~10^{-12} that of normal resistance) at 100kHz and only 10^{-6} of the total current is carried by normal electrons

But……

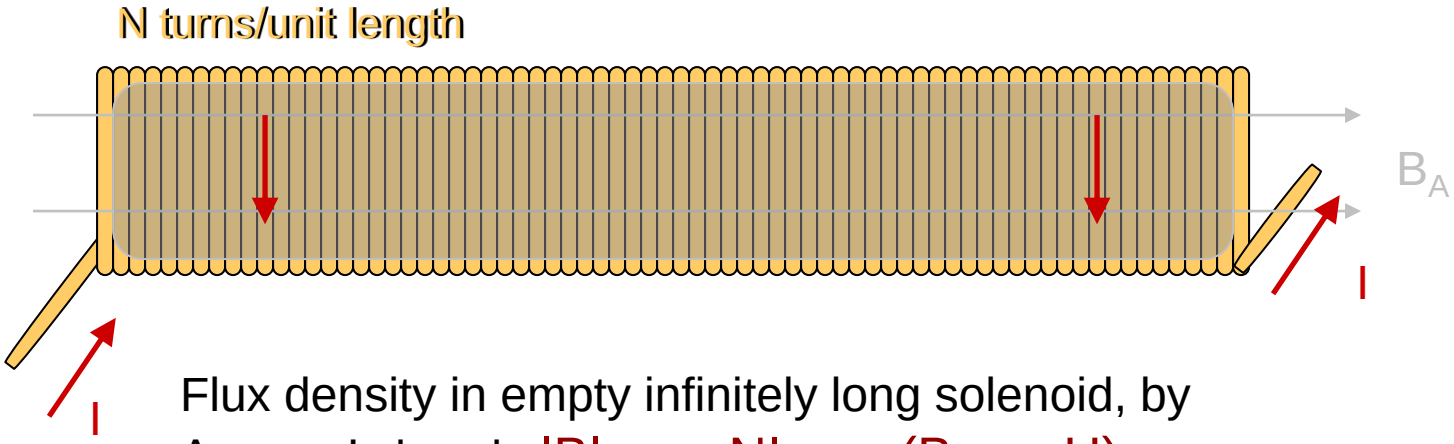
At higher (optical) frequencies (~10^{11}Hz) the superconductor appears entirely normal

…..to be discussed later!!

Some definitions

In free space: $\oint \underline{H} \cdot \underline{dl} = I$ $\underline{H}$ is “magnetic field” in A/m

$\oint \underline{B} \cdot \underline{dl} = \mu_o I$ $\underline{B}$ is magnetic flux density measured in Tesla



Flux density in empty infinitely long solenoid, by Ampere’s law, is $|\underline{B}| = \mu_o NI$ $(\underline{B} = \mu_o \underline{H})$

Flux density in solenoid containing infinitely long sample with a magnetisation per unit volume of $\underline{M_v}$ is

$\underline{B} = \mu_o(\underline{H} + \underline{M_v})$ ($\underline{M_v}$ has units of A/m)

Susceptibility

For most materials (except ferromagnets, and paramagnets in very high magnetic fields and low temperatures)

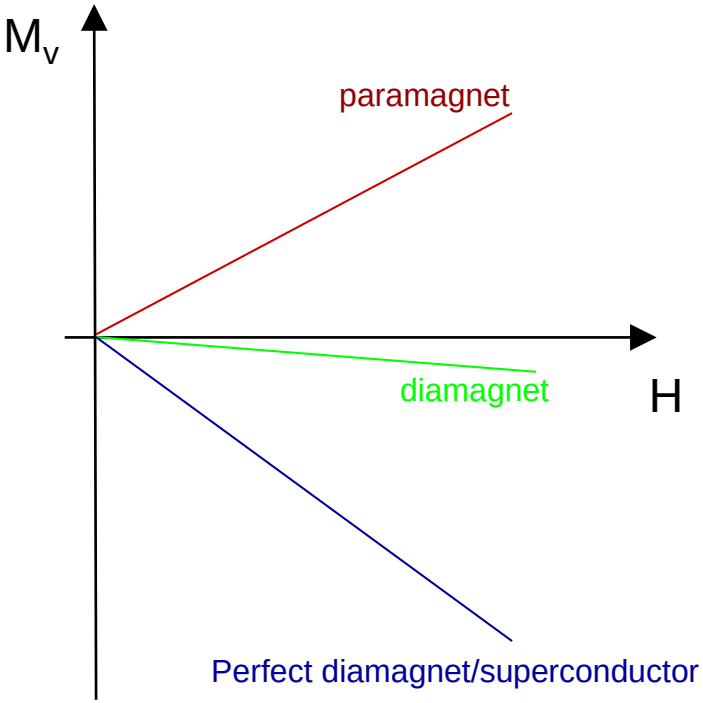
$\underline{M_v} \propto \underline{H}$ with $\underline{M_v} = \chi \underline{H}$

where χ is the (dimensionless) susceptibitity

so: $\underline{B} = \mu_o \underline{H}(1 + \chi)$

For most paramagnetic materials $\chi \sim 10^{-3}$,
for diamagnets $\chi \sim -10^{-5}$

If a superconductor always maintains
$B=0$ within its interior, then $\chi = -1$



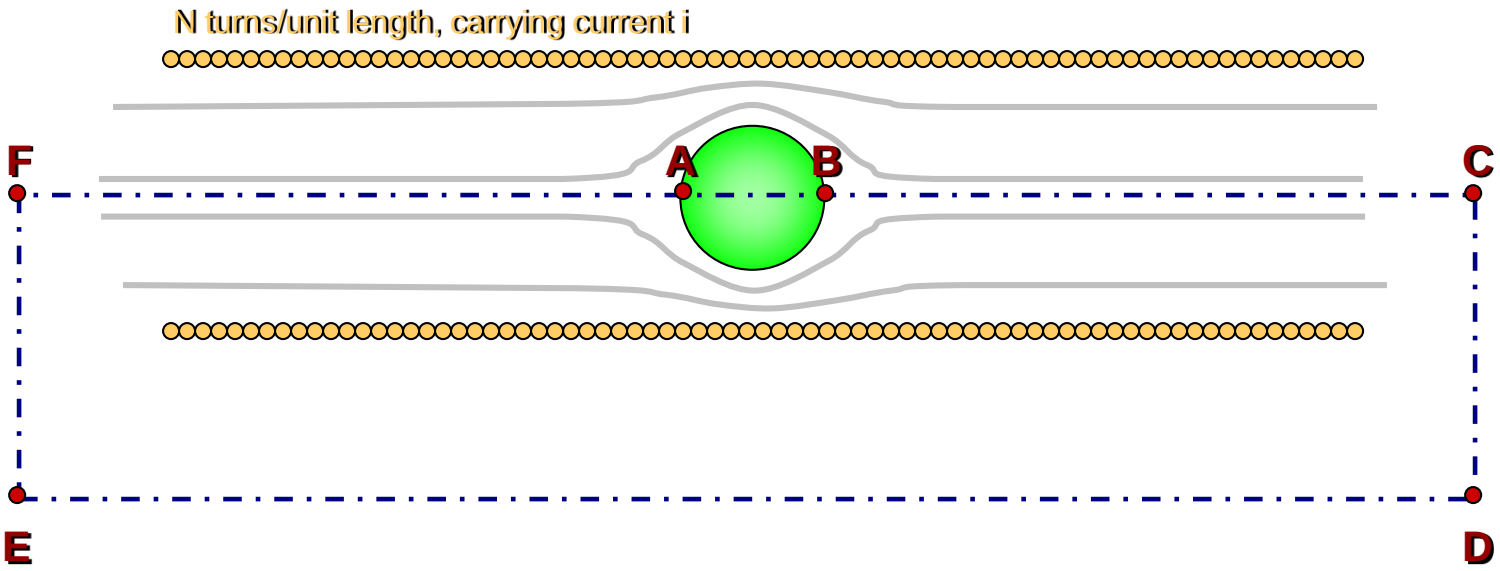
A superconductor can therefore be described as

(a) a “perfect diamagnet”

or (b) having screening currents flowing at the surface producing a field of magnitude M_V equal and opposite to H

Note that B=0 but H $\neq$ 0 within the superconductor

Demagnetisation



With a superconducting sample in the solenoid

Around ABCDEF $\oint \underline{H} \cdot \underline{dl} = Ni$ and $\oint \underline{H} \cdot \underline{dl} = \int_{AB} \underline{H_i} \cdot \underline{dl} + \int_{BCDEF} \underline{H_e} \cdot \underline{dl}$ (1)

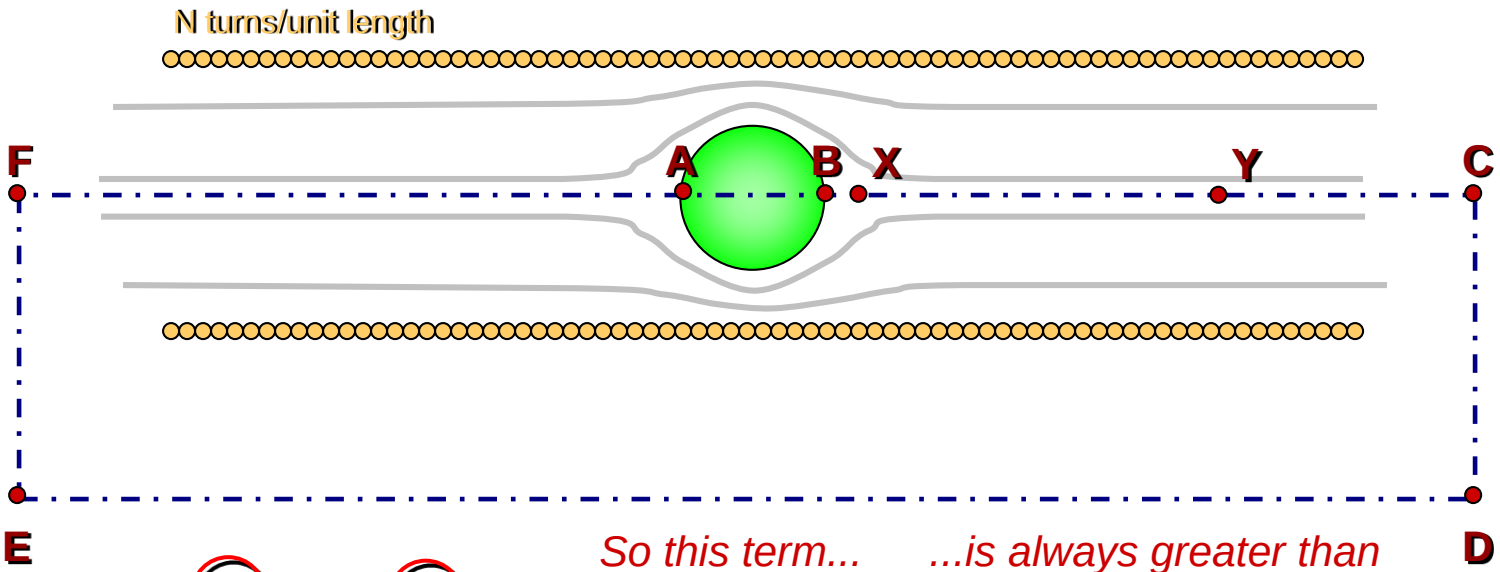
But with the sample removed from the solenoid

Around ABCDEF $\oint \underline{H} \cdot \underline{dl} = Ni$ and $\oint \underline{H} \cdot \underline{dl} = \int_{AB} \underline{H_a} \cdot \underline{dl} + \int_{BCDEF} \underline{H'_e} \cdot \underline{dl}$ (2)

H_a = field applied to sample,
H'_e = external field without sample

H_i = internal field within sample,
H_e= external field with sample

Demagnetisation



Equating (2) and (1)

So this term... *...is always greater than or equal to this term*

$$\int_{AB} \underline{H_a} \cdot \underline{dl} + \int_{BCDEF} \underline{H'_e} \cdot \underline{dl} = \int_{AB} \underline{H_i} \cdot \underline{dl} + \int_{BCDEF} \underline{H_e} \cdot \underline{dl}$$

At **X,** screening currents cause H_e to be less than H_e'

But at **Y,** the effects of the screening currents are negligible, and $H_e = H_e$'

Therefore and $H_i \geq H_a$ and the field inside the superconductor can exceed the applied field!

Demagnetisation corrections

In general we write

$$H_i = H_a - H_D$$

For the special case of an ellipsoid, the field is uniform throughout the body and

$$H_i = H_a - nM_v$$

where n is the demagnetising factor

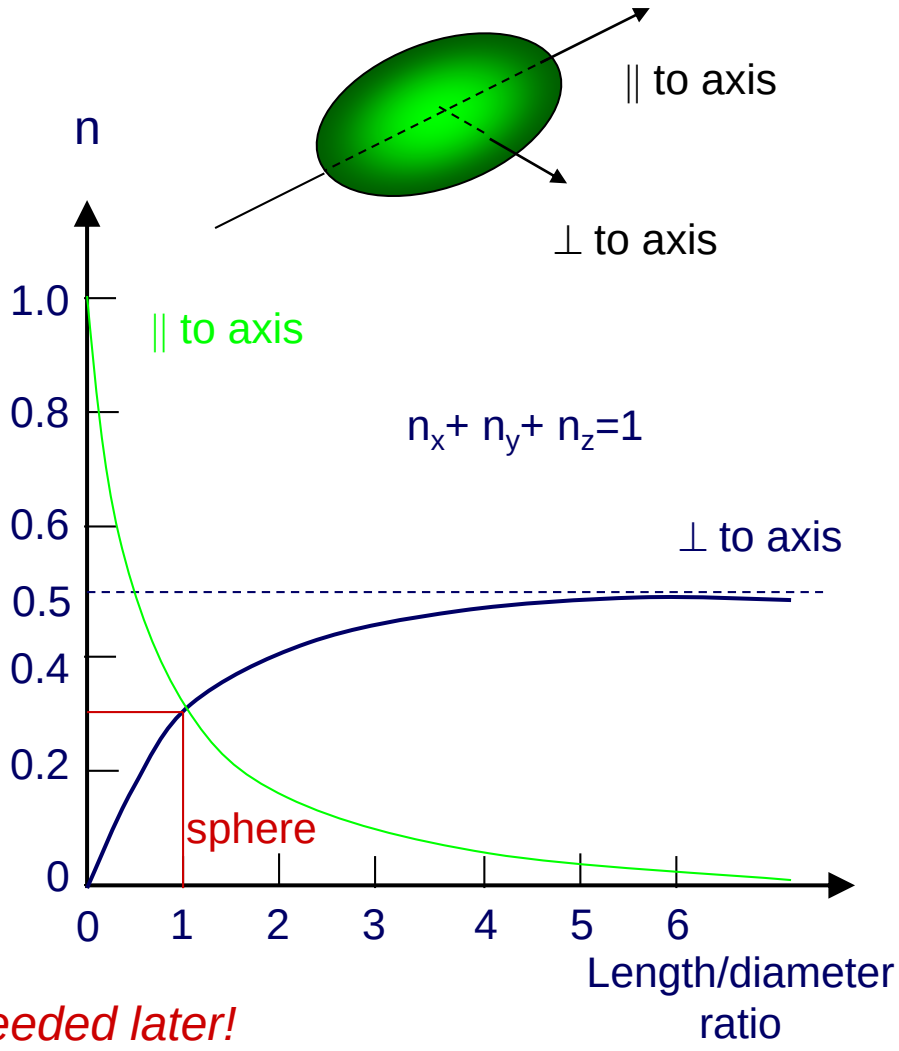
For a superconductor $M_v < 0$, so $H_i > H_a$

or $M_v = \chi H_i = -H_i$

so $H_i (1-n) = H_a$

and

$$H_i = \frac{H_a}{(1-n)}$$



This will be needed later!

The London Model

An important consequence of flux exclusion in superconductors is that

If magnetic flux density must remain zero in the bulk of a superconductor, then any currents flowing through the superconductor can flow only at the surface

However a current cannot flow entirely at the surface or the current density would be infinite

The concept of “penetration depth” must be introduced

In 1934 F and H London proposed a macroscopic phenomenological model of superconductivity based upon the two-fluid model

The London model introduced the concept of the (London) penetration depth and described the Meissner effect by considering superconducting electrodynamics



Some electrodynamics

Consider a perfect conductor in which the current is carried by n electrons

current density $\underline{J} = ne\underline{v}$ 1

and in an electric field $m\dot{\underline{v}} = e\underline{E}$ 2

so the rate of increase of current density is $\dot{\underline{J}} = ne^2\underline{E}/m$ 3

Maxwell's equations give $\mathrm{curl}\underline{H} = \underline{J} + \dot{\underline{D}}$ 4

and $\dot{\underline{B}} = -\mathrm{curl}\underline{E}$ 5

Assuming the displacement current $\dot{\underline{D}} = 0$ and $(1+\chi) \approx 1$

equation 4 gives $\mathrm{curl}\underline{B} = \mu_o \underline{J}$ 6

Then 5 and 3 give $\dot{\underline{B}} = -m\,\mathrm{curl}\dot{\underline{J}}/ne^2$ 7

and 6 and 7 give $\dot{\underline{B}} = -\dfrac{m}{\mu_o ne^2}\mathrm{curl}\,\mathrm{curl}\dot{\underline{B}}$ 8

Some more electrodynamics

$\dot{\underline{B}} = -\frac{m}{\mu_o n e^2}\,\mathrm{curl}\,\mathrm{curl}\,\dot{\underline{B}}$ can be simplified using the standard identity

$$\mathrm{curl}\,\mathrm{curl}\,\dot{\underline{B}} = \mathrm{grad}\,\mathrm{div}\,\dot{\underline{B}} - \nabla^2\dot{\underline{B}} \qquad (\mathrm{div}\,\underline{B} = 0,\ \mathrm{div}\,\dot{\underline{B}} = 0)$$

So equation 8 becomes $\boxed{\nabla^2\dot{\underline{B}} = \frac{\dot{\underline{B}}}{\alpha}}$ with $\alpha = \frac{m}{\mu_o n e^2}$

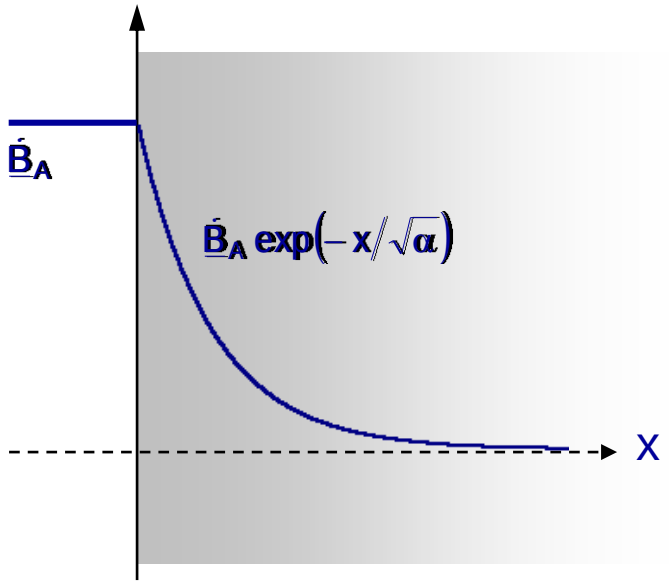
In one dimension this is simply

$$\frac{\partial^2\dot{\underline{B}}}{\partial x^2} = \frac{\dot{\underline{B}}}{\alpha}$$

To which the solution is

$$\dot{\underline{B}}(x) = \dot{\underline{B}}_A \exp\left(-x/\sqrt{\alpha}\right)$$

So, for $x \gg \sqrt{\alpha}$ ~10^{-6}cm, ie inside a perfect conductor B does not change ($\dot{\underline{B}}(x) = 0$) when B_A changes



$\dot{\underline{B}}$ decreases exponentially when we move into the material

The London penetration depth

Experiment had shown that not only $\dot{\underline{B}} = 0$ but also $\underline{B} = 0$ within a superconductor

F and H London suggested that not only $\nabla^2 \dot{\underline{B}} = \frac{\dot{\underline{B}}}{\alpha}$

but also $\nabla^2 \underline{B} = \frac{\underline{B}}{\alpha}$

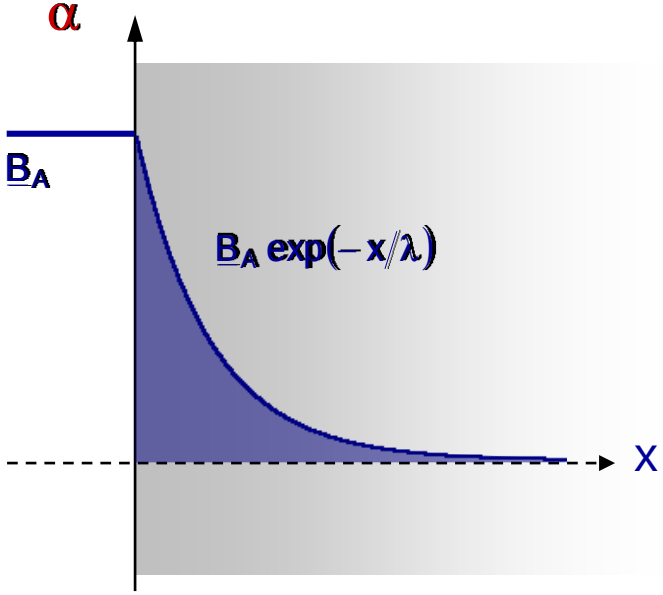
To which the solution is

$$\underline{B}(x) = \underline{B}_A \exp(-x/\lambda_L)$$

where $\lambda_L = \sqrt{\alpha} = \sqrt{m/\mu_o n_s e^2}$

λ_L is known as the London penetration depth

It is a fundamental length scale of the superconducting state



Surface currents

Working backwards from the London equation $\nabla^2 \underline{B} = \frac{\underline{B}}{\alpha}$ to equation 7

gives $\underline{B} = -m\,\mathrm{curl}\,\underline{J}_s / n_s e^2$

So, for a uniform field parallel to the surface (z-direction) the "new" equation 6 becomes

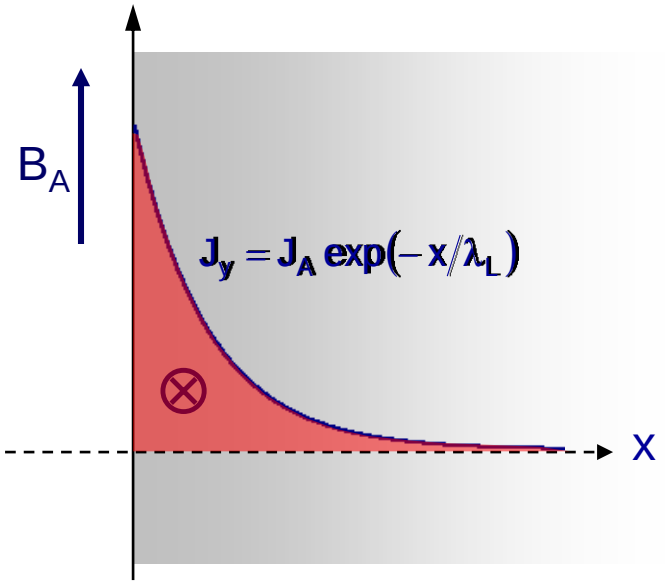
$$-\frac{\partial B}{\partial x} = \mu_o J_y$$

and as $\frac{\partial B}{\partial x} = -\frac{B_A}{\lambda_L}\exp(-x/\lambda_L)$

$$J_y = -\frac{B_A}{\mu_o \lambda_L}\exp(-x/\lambda_L)$$

or $J_y = \underline{J}_A \exp(-x/\lambda_L)$

So current flows not just at the surface, but within a penetration depth λ_L



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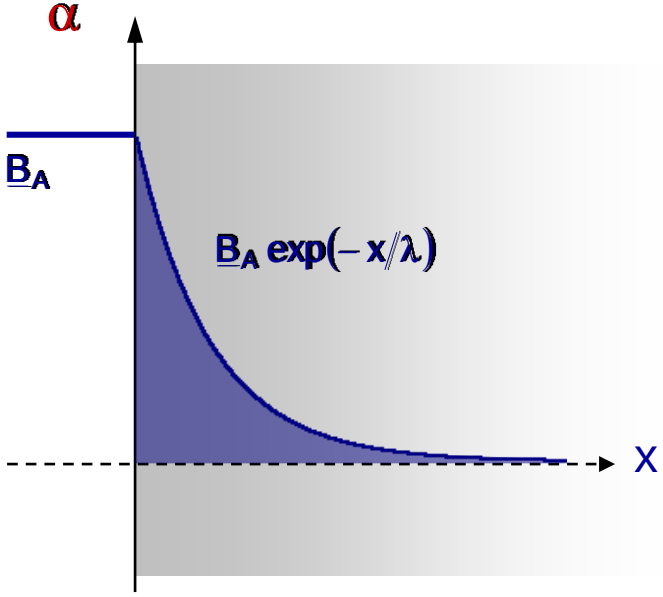
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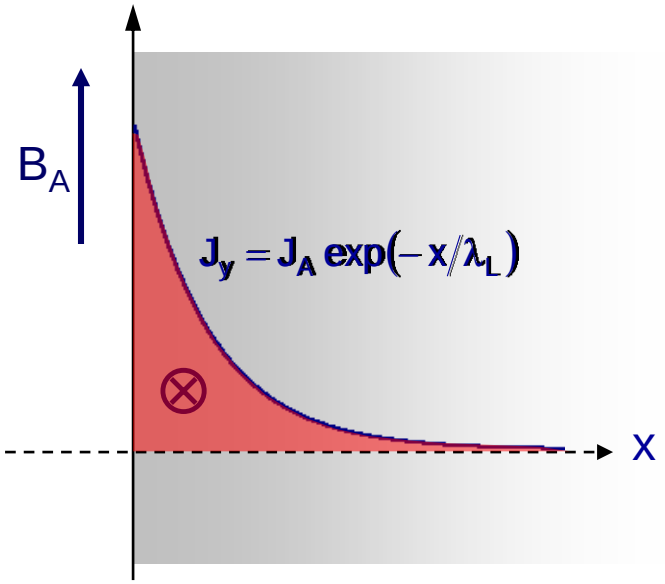
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So current flows not just at the surface, but within a penetration depth λ_L



The London model - a summary

The Londons produced a phenomenological model of superconductivity which provided equations which described *but did not explain* superconductivity

Starting with the observation that superconductors expel all magnetic flux from their interior, they demonstrated the concept of the Penetration Depth, showing that

- Flux does penetrate, but falls of exponentially on a length scale, λ
- Electric current flows only at the surface, again falling off exponentially on a length scale, λ

So, in just one dimension we have

$$\underline{B}(x) = \underline{B}_A \exp(-x/\lambda_L)$$

and

$$J_y(x) = J_A \exp(-x/\lambda_L)$$

with $\lambda_L = \sqrt{m/\mu_o n_s e^2}$ the London penetration depth

Critical fields

Onnes soon found that the normal state of a superconductor could be recovered by applying a magnetic field greater than a critical field, $H_c = B_c/\mu_o$

This implies that above H_c the free energy of the normal state is lower than that associated with the superconducting state

The free energy per unit volume of the superconductor in zero field is

$G_S(T, 0)$ below T_c and $G_N(T,0)$ above T_c

The change in free energy per unit volume associated with applying a field H_a parallel to the axis of a rod of superconductor (so as to minimise demagnetisation) is

$$\Delta G(H_a) = -\mu_o \int_0^{H_a} M_v \, dH_a$$

where M_v is the volume magnetisation

For most magnetic materials M_v is positive so the free energy is lowered when a field is applied, but if M_v is negative, the free energy increases

but for a superconductor M_V is negative...

Critical fields

We have $G_S(T,H_a) = G_S(T,0) - \mu_o \int_0^{H_a} M_v dH_a$

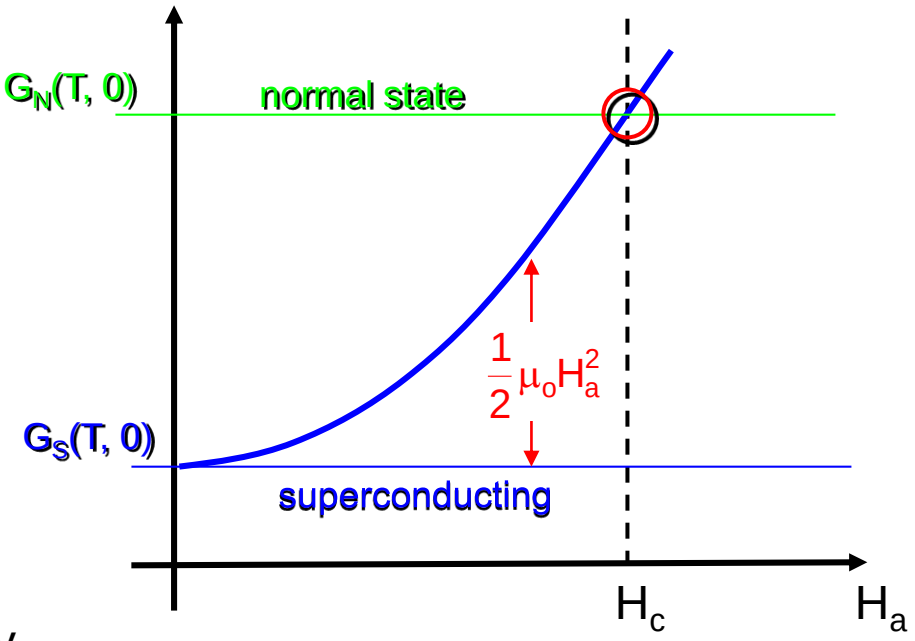
So, in the absence of demagnetising effects, $M_V = \chi H_a = -H_a$, and

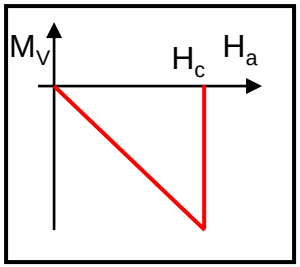
$$G_S(T,H_a) = G_S(T,0) + \mu_o \int_0^{H_a} H_a dH_a$$

$$= G_S(T,0) + \frac{1}{2}\mu_o H_a^2$$

When the magnetic term in the free energy is greater than $G_N(T, 0)-G_S(T,0)$ the normal state is favoured, ie

$$H_c(T) = \left[\frac{2}{\mu_o}(G_N(T,0) - G_S(T,0))\right]^{1/2}$$

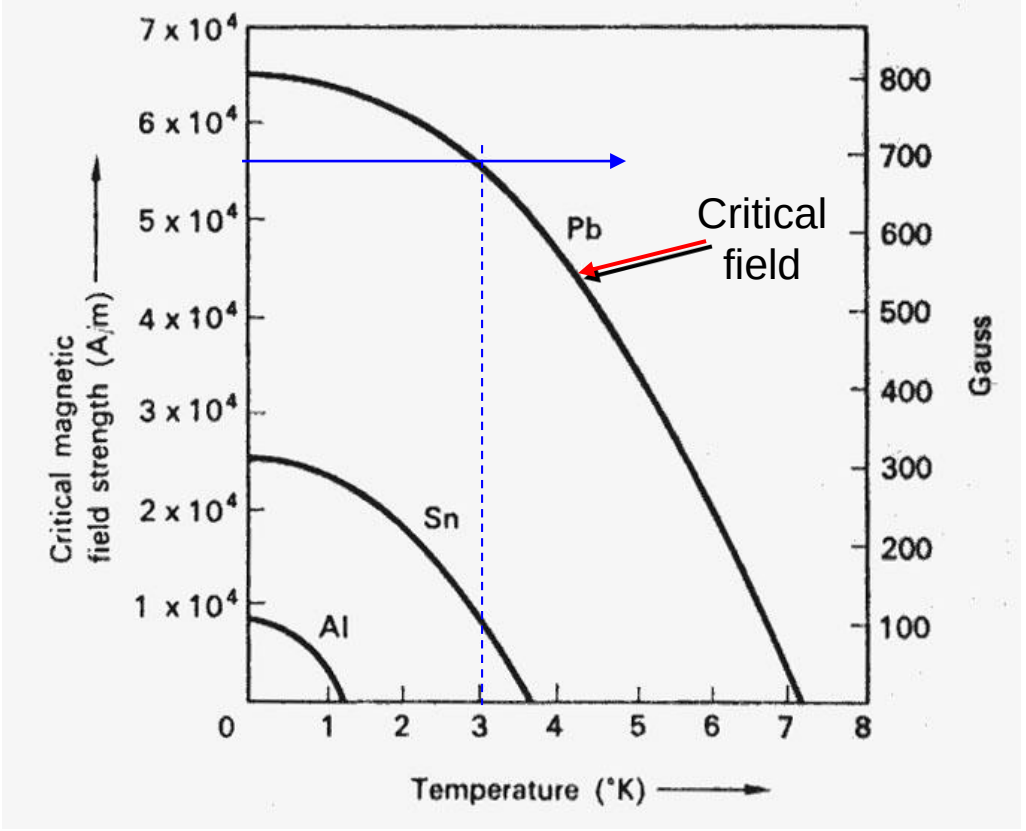




Critical fields - temperature dependence

Experimentally it is found that

$$H_c(T) = H_o\left\{1-(T/T_c)^2\right\}$$



Critical currents

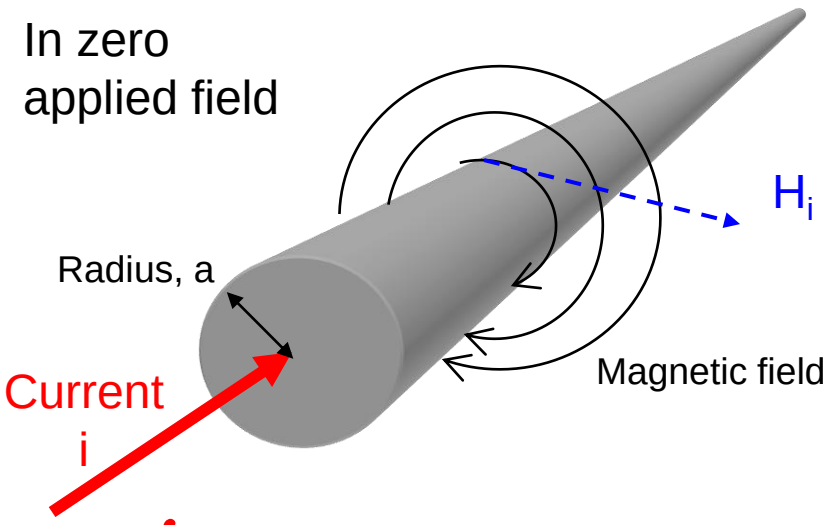
If a superconductor has a *critical magnetic field, H_c*, one might also expect a *critical current density, J_c.*

The current flowing in a superconductor can be considered as the sum of the transport current, J_i, and the screening currents, J_s.

If the sum of these currents reach J_c then the superconductor will become normal.

The larger the applied field, the smaller the transport current that can be carried and *vice versa*

J_c has a similar temperature dependence to H_c, and T_c is similarly lowered as J increases



$$\oint \underline{H_i}.\underline{dl} = i \quad \text{so} \quad 2\pi a H_i = i$$

and $i_c = 2\pi a H_c$

The critical current density of a long thin wire in zero field is therefore

$$j_c = \frac{2\pi a H_c}{\pi a^2} \quad \text{ie} \quad \boxed{j_c = \frac{2H_c}{a}}$$

Typically $j_c \sim 10^6 A/m^2$ for type I superconductors

The intermediate state

A conundrum:

If the current in a superconducting wire of radius a just reaches a value of

$$i_c = 2\pi a H_c$$

the surface becomes normal leaving a superconducting core of radius a'<a

The field at the surface of this core is now

$$H' = i_c / 2\pi a' \; > \; H_c$$

So the core shrinks again - and so on until the wire becomes completely normal

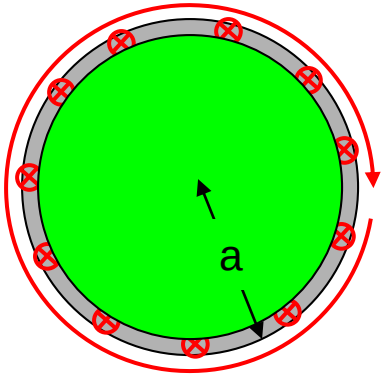
But
- when the wire becomes completely normal the current is uniformly distributed across the full cross section of the wire

Taking an arbitrary line integral around the wire, say at a radius a'<a, now gives a field that is smaller than H_c as it encloses a current which is much less than i_c

….so the sample can become superconducting again!

and the process repeats itself ……..this is of course unstable

….schematically



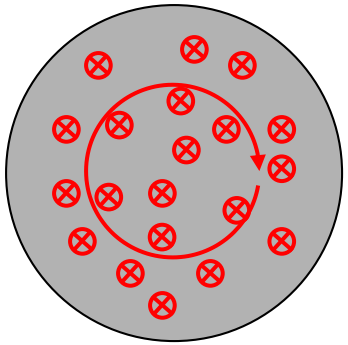
Critical current is reached when the line integral of the field around the loop is

$i_c = 2\pi a H_c$

Current density is

$j_c = i/\pi a^2$.

Note current flowing within penetration depth of the surface. Superconducting state collapses



The sample is normal and current is distributed uniformly over cross section.

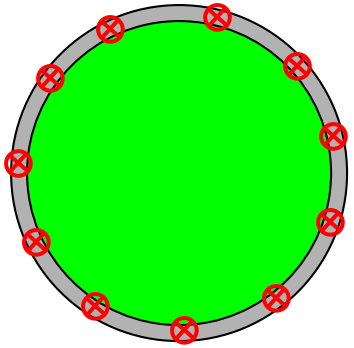
Current enclosed by loop at radius a'<a is

$i = i_c a'^2/a^2 < i_c$

also the line integral of the field around the loop gives

$H = i/2\pi a' = i_c a'/2\pi a^2 < H_c$

…..so the sample can become superconducting again

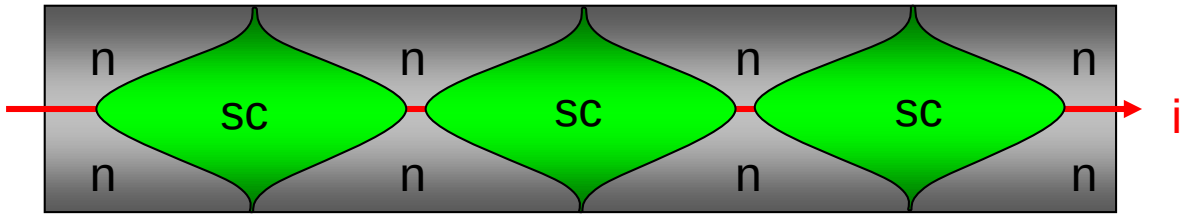


…..and the process repeats itself

The intermediate state

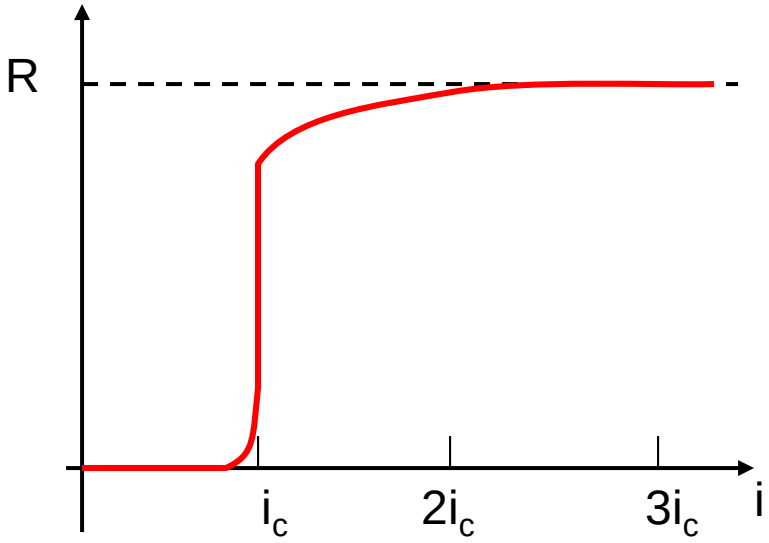
Instead of this unphysical situation the superconductor breaks up into regions, or domains, of normal and superconducting material

The shape of these regions is not fully understood, but may be something like:



The superconducting wire will now have some resistance, and some magnetic flux can enter

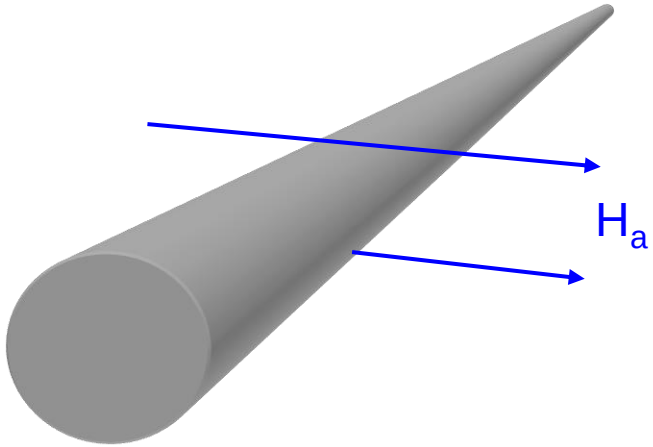
Moreover, the transition to the normal state, as a function of current, is not abrupt



Field-induced intermediate state

A similar state is created when a superconductor is placed in a magnetic field:

Consider the effect of applying a magnetic field perpendicular to a long thin sample



The demagnetising factor is $n=0.5$ in this geometry, so the internal field is

$$H_i = H_a/(1-n) = 2H_a$$

So, the internal field reaches the critical field when

$$H_a = H_c/2$$

The sample becomes normal
- so the magnetisation and hence demagnetising field falls to zero

The internal field must now be less than H_c (indeed it is only $H_c/2$)

The sample becomes superconducting again, and the process repeats

Again this is unphysical

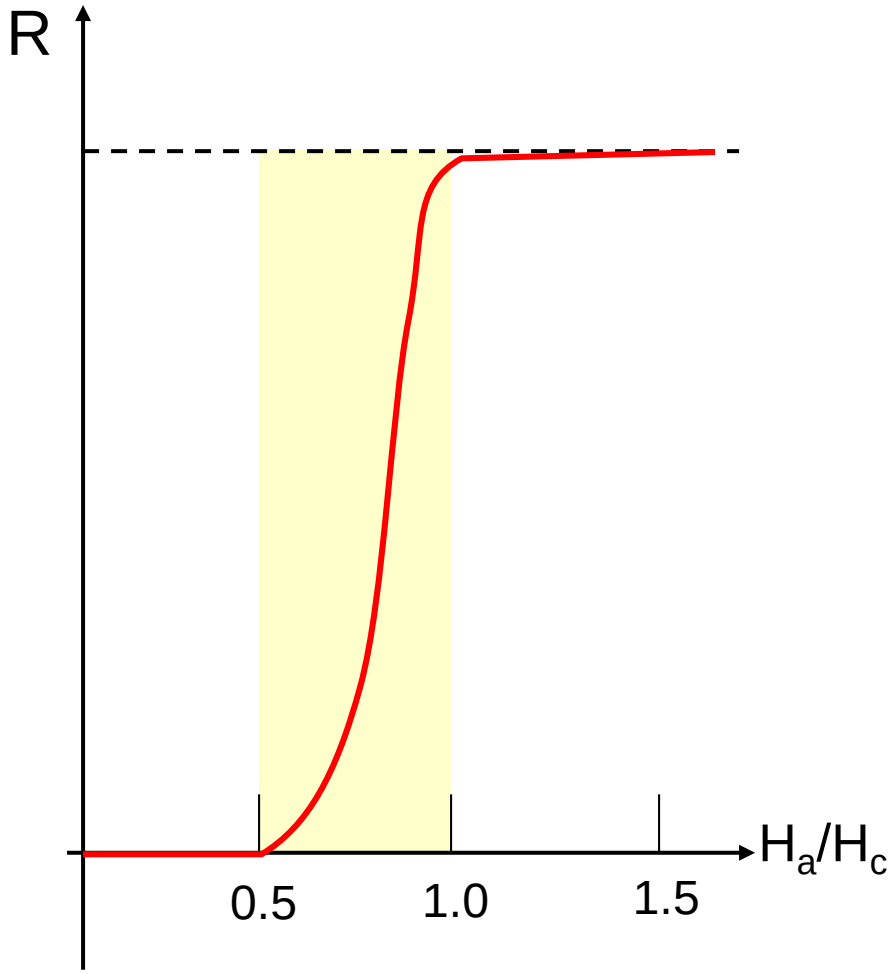
Field-induced intermediate state

Once again the superconductor is stabilised by breaking down into normal and superconducting regions

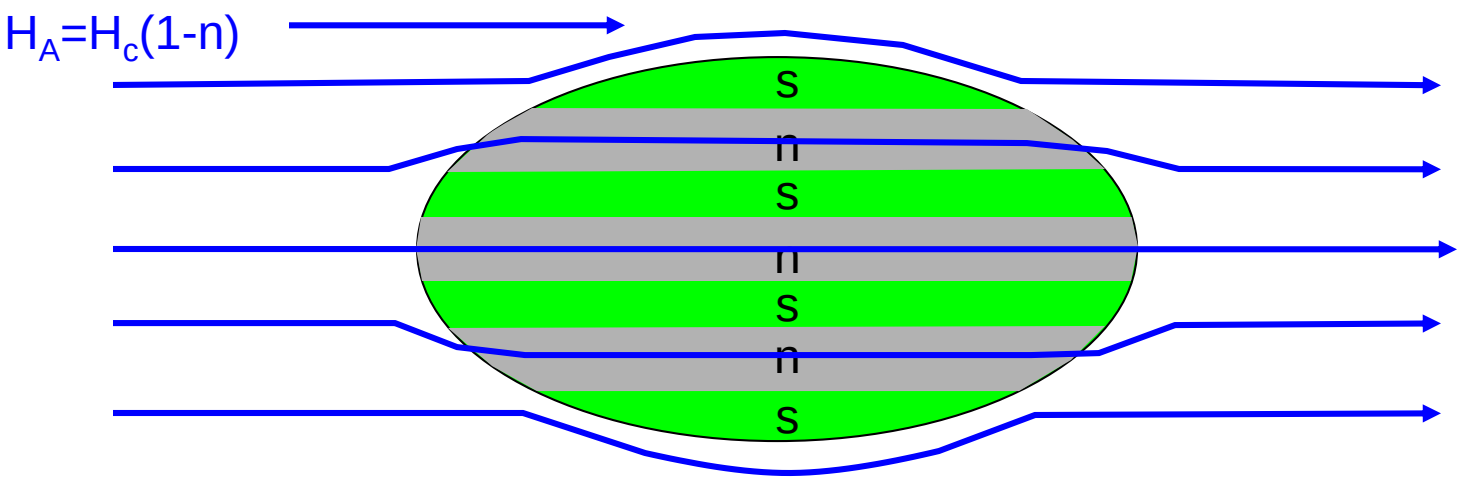
Resistance begins to return to the sample at applied fields well below H_c - but at a value that *depends upon the shape of the sample* through the demagnetising factor n

When the field is applied perpendicular to the axis of a long thin sample n=0.5, and resistance starts to return at $H_a = H_c/2$

For this geometry the sample is said to be in the ***intermediate state*** between $H_a = H_c/2$ and $H_a = H_c$



Field distribution in the inetrmediate state



When $H_i=H_c$ the sample splits into normal and superconducting regions which are in equilibrium for $H_c(1-n)<H_a<H_c$

$B_\perp$ at the boundaries must be continuous, and $B=0$ within superconducting region, so $B_\perp=0$ in both superconducting and normal regions

- *the boundaries must be parallel to the local field*

H must also be parallel to the boundary, and $H_{||}$ must also be continuous at the boundary, therefore H must be the same on both sides of the boundary

On the normal side $H_i=H_c$, so on the superconducting side $H_i=H_c$

Therefore a stationary boundary exists only when $H_i=H_c$

The Intermediate state

A thin superconducting plate of radius a and thickness t with a>>t has a demagnetising factor of

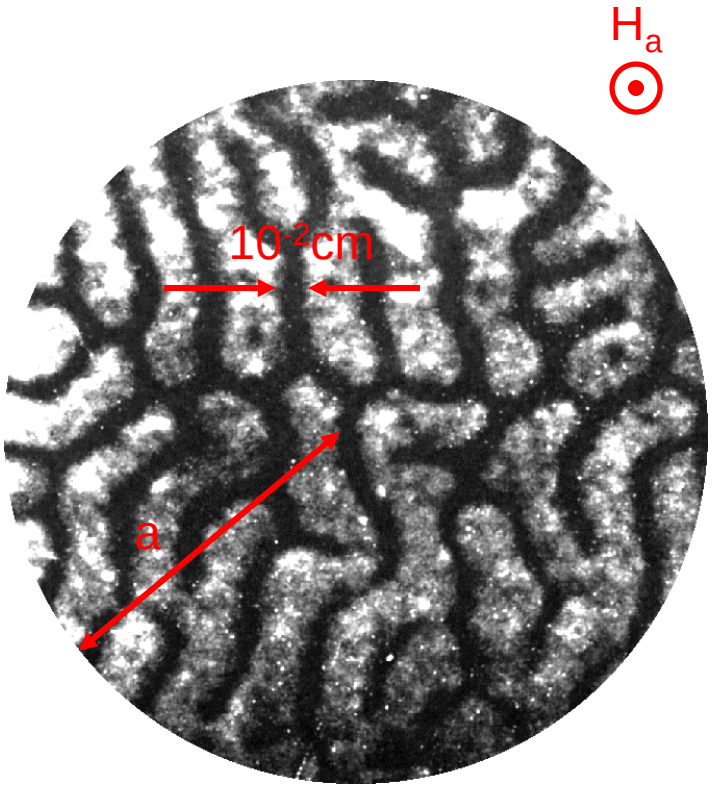
$$n \approx 1 - t/2a$$

So, with a field H_a applied perpendicular to the plate the internal field is

$$H_i = H_a/(1-n) \quad = 2a.H_a/t$$

and only a very small applied field is needed to reach $H_a = H_c$

Generally, for elemental superconductors the superconducting domains are of the order of 10^{-2} to 10^{-1} cm thick, depending upon the applied field



The dark lines are superconducting regions of an aluminium plate "decorated" with fine tin particles

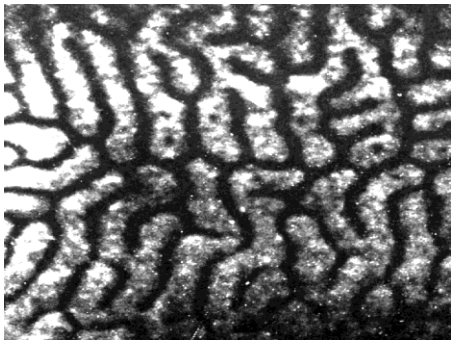
Surface energy

The way a superconductor splits into superconducting and normal regions is governed by the *surface energy* of the resulting domains:

Surface energy >0 **Type I**

The free energy is minimised by minimising the total area of the interface

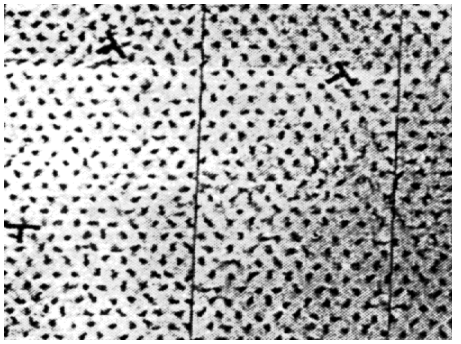
hence relatively few thick domains



Surface energy <0 **Type II**

Energy is released on formation of a domain boundary

hence a large number of thin domains



In the second case it is energetically favourable for the superconductor to spontaneously split up into domains even in the absence of demagnetising effects

To understand this we need to introduce the concept of the "coherence length"

The Pippard coherence length

In 1953 Sir Brian Pippard considered

1. N/S boundaries have positive surface energy
2. In zero magnetic field superconducting transitions in pure superconductors can be as little as 10^{-5}K wide

   Thus all electrons in the sample must participate in superconductivity and there must be long range order or coherence between the electrons
3. Small particles of superconductors have penetration depths greater than those of bulk samples

   Therefore superconducting electron densities must change at a relatively slow rate through the sample

He concluded:

The superconducting electron density n_s cannot change rapidly with position...

….it can only change appreciably of a distance of $\xi \sim 10^{-4}$cm,

The boundary between normal and superconducting regions therefore cannot be sharp….

…..n_s has to rise from zero at the boundary to a maximum value over a distance ξ

ξ is the Pippard coherence length

The Pippard coherence length

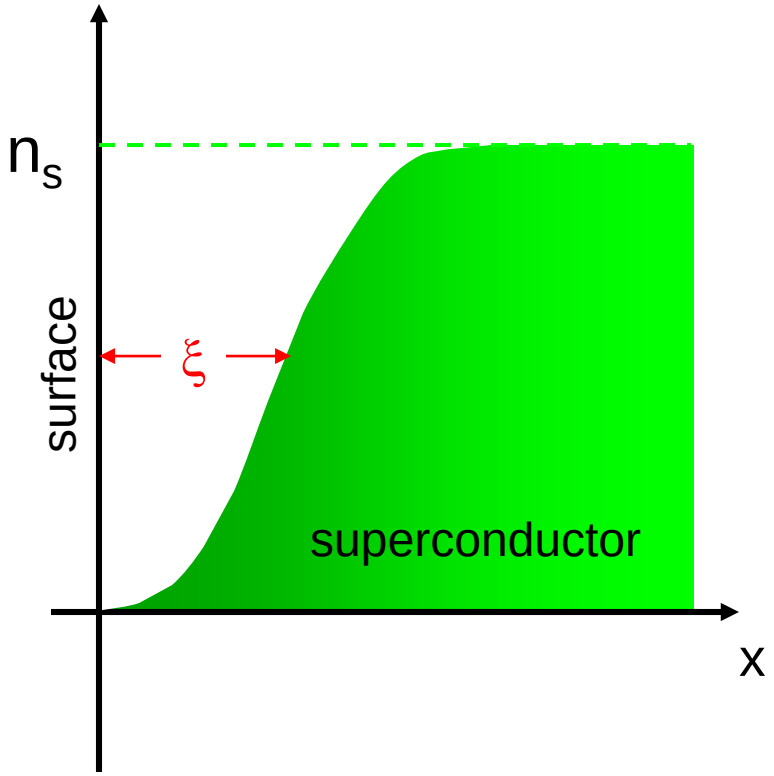
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ξ is the Pippard coherence length



Surface energy considerations

We now have two fundamental length scales of the superconducting state:

The penetration depth, λ, is the length scale over which magnetic flux can penetrate a superconductor

The coherence length, ξ, is the length scale over which the superelectron density can change

We also know that the superconducting region is "more ordered" than the normal region so that

$$g_n - g_s = \frac{1}{2}\mu_o H_c^2$$

which changes on the length scale of ξ

1

Whilst in a magnetic field the superconductor acquires a magnetisation to cancel the internal flux density, hence deep in the material

$$g_s - g_n = \frac{1}{2}\mu_o H_c^2$$

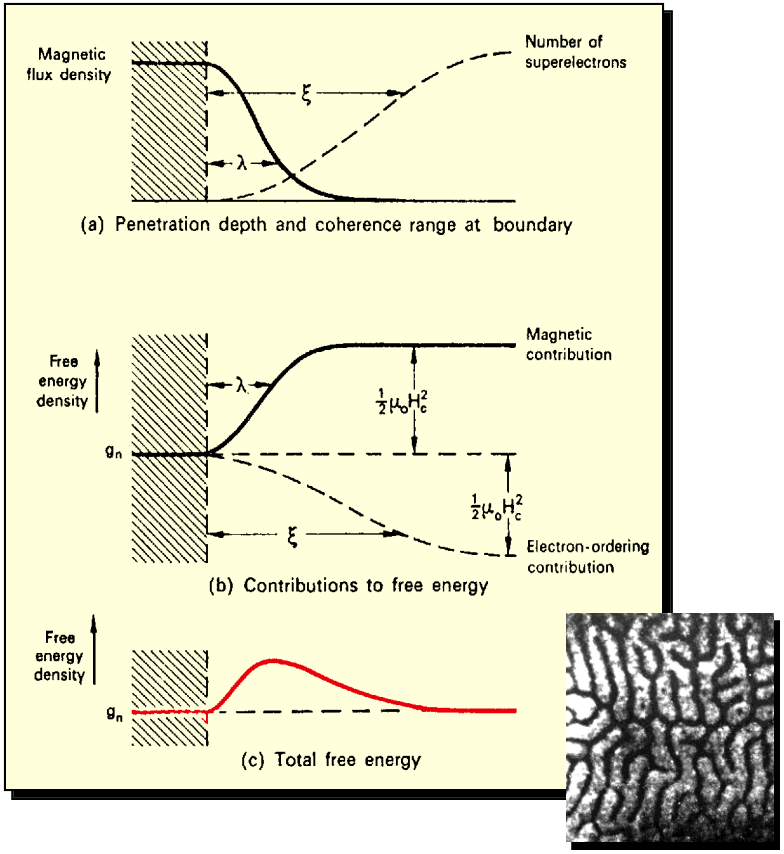
which changes on the length scale of λ

2

Deep inside the superconductor these free energy terms (1 and 2) cancel exactly, but what happens closer to the surface?

Positive and negative surface energy

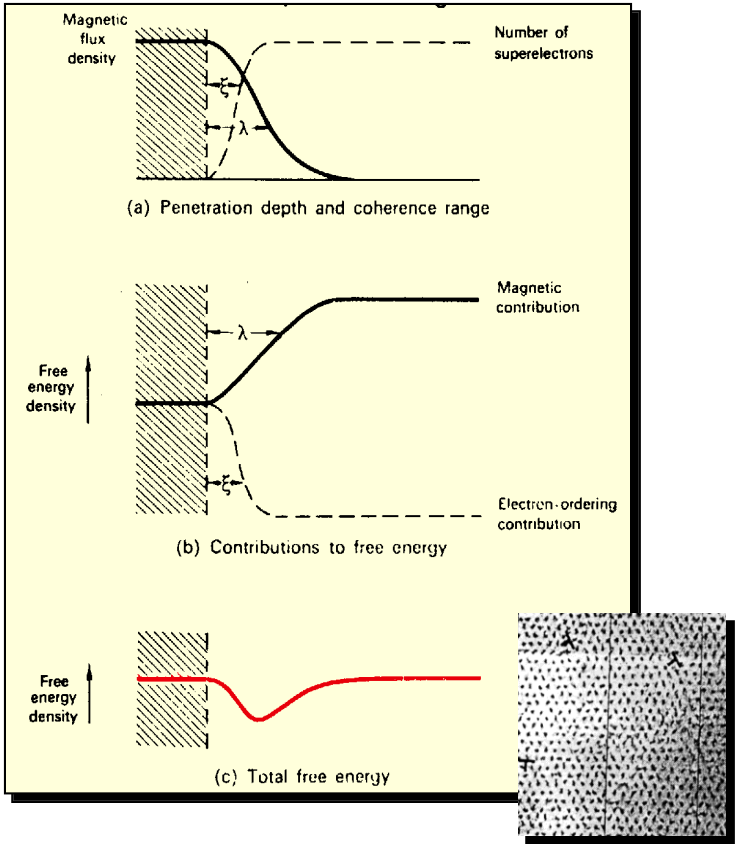
For $\xi > \lambda$



Surface energy is positive:

Type I superconductivity

For $\xi < \lambda$



Surface energy is negative:

Type II superconductivity

Conditions for Type II Superconductivity

If the surface energy is negative we expect Type II superconductivity

Normal “cores” ,“flux lines” or “vortices” will appear and arrange themselves into an hexagonal lattice due to the repulsion of the associated magnetic dipoles

A normal core increases the free energy per unit length of core by an amount

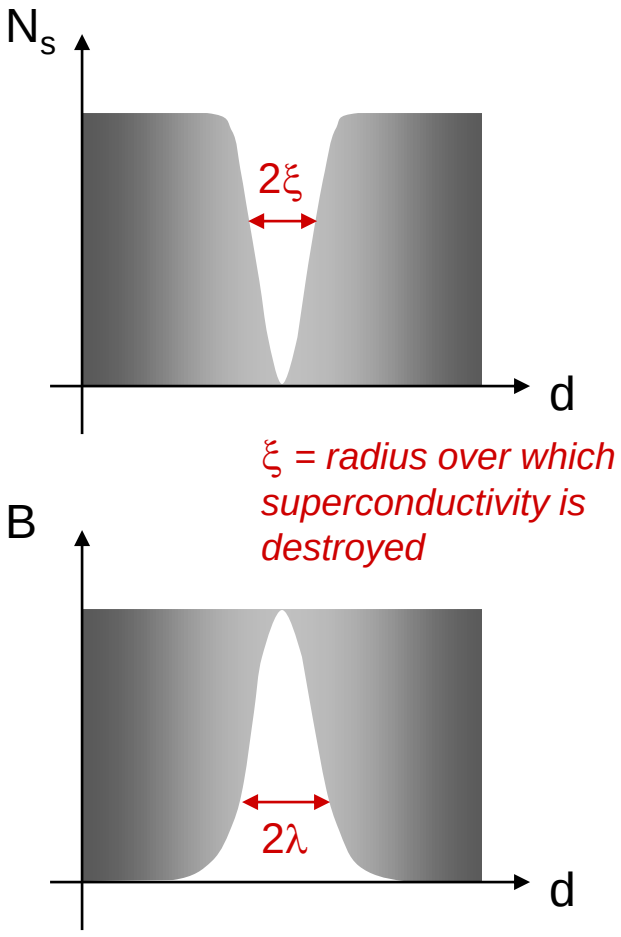
$$\pi\xi^2 \frac{1}{2}\mu_o H_c^2$$

…but over a length scale λ the material is not fully diamagnetic so in a field H_a there is a local decrease in magnetic energy of

$$\pi\lambda^2 \frac{1}{2}\mu_o H_a^2$$

So for a net reduction of energy

$$\pi\xi^2 \frac{1}{2}\mu_o H_c^2 < \pi\lambda^2 \frac{1}{2}\mu_o H_a^2$$



ξ = radius over which superconductivity is destroyed

λ = radius of vortex

The Lower and Upper Critical Fields

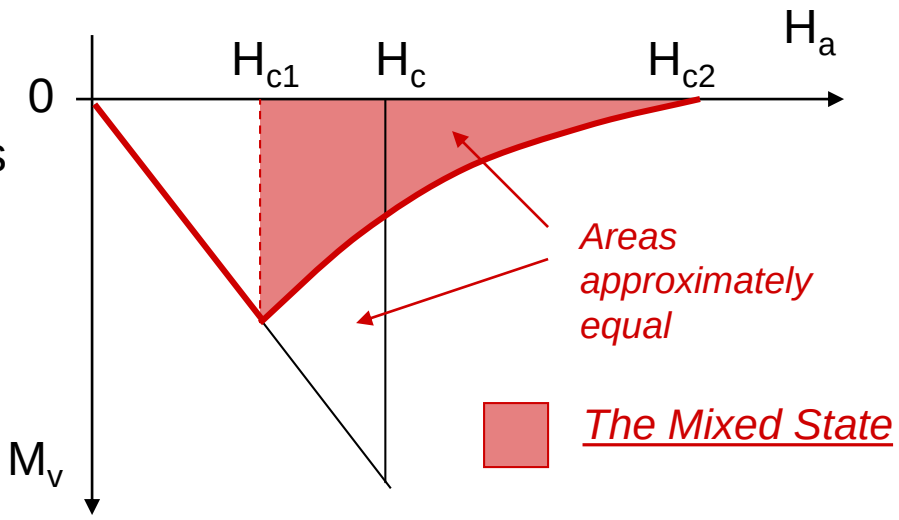
$$\pi\xi^2 \tfrac{1}{2}\mu_o H_c^2 < \pi\lambda^2 \tfrac{1}{2}\mu_o H_a^2$$

Therefore magnetic cores or flux lines will spontaneously form for

$$\lambda^2 H_a^2 > \xi^2 H_c^2$$

ie, if $$H_a = H_{c1} > \frac{\xi}{\lambda} H_c$$

providing $$\frac{\lambda}{\xi} = \kappa > 1$$



where κ is the Ginzburg-Landau parameter

H_{c1} is known as the lower critical field

(A more rigorous G-L treatment shows κ must be greater than $\sqrt{2}$ *-see later lectures*)

As some magnetic flux has entered the sample it has lower free energy than if it was perfectly diamagnetic, therefore a field greater than H_c is required to drive it fully normal

This field, H_{c2}, is the upper critical field.

Note: for Nb, $\kappa \sim 1$

Ginzburg-Landau Theory

Everything we have considered so far has treated superconductivity semi-classically

However we know that superconductivity must be a deeply quantum phenomenon

In the early 1950s Ginzburg and Landau developed a theory that put superconductivity on a much stronger quantum footing

Their theory, which actually predicts the existence of Type II superconductivity, is based upon the general Landau theory of "second order" or "continuous" phase transitions

In particular they were able to incorporate the concept of a spatially dependent superconducting electron density n_s, and allowed n_s to vary with external parameters

Note: in the London theory $\lambda = \sqrt{m/\mu_o n_s e^2}$ but n_s does not depend upon distance as the Pippard model demands. The concept of coherence length is entirely absent.

Landau Theory of Phase Transitions

As a reminder of Landau theory, take the example of a *ferromagnetic to paramagnetic transition* where the free energy is expressed as

$$F(M) = F(0) + a(T)M^2 + bM^4 \qquad F(M) = F(0) + a_0(T - T_{CM})M^2 + bM^4$$

M is the magnetisation - the so-called *order parameter* of the magnetised ferromagnetic state

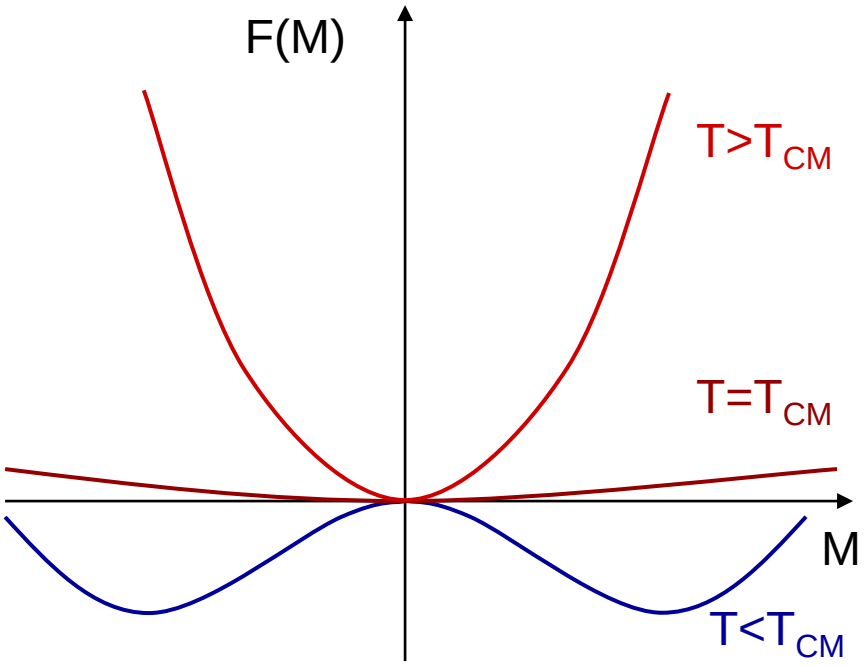
The stable state is found at the minimum of the free energy, ie when

$$\frac{\partial F(T,M)}{\partial M} = 0$$

We find M=0 for $T>T_{CM}$

M≠0 for $T<T_{CM}$

Any second order transition can be described in the same way, replacing M with an order parameter that goes to zero as T approaches T_C



The Superconducting Order Parameter

We have already suggested that superconductivity is carried by superelectrons of density n_s

n_s could thus be the "order parameter" as it goes to zero at T_c

However, Ginzburg and Landau chose a quantum mechanical approach, using a wave function to describe the superelectrons, ie

$$\Psi(\underline{r}) = |\Psi(\underline{r})| e^{i\varphi(\underline{r})}$$

This complex scalar is the Ginzburg-Landau order parameter

(i) its modulus $|\Psi^* \Psi|$ is roughly interpreted as the number density of superelectrons at point $\underline{r}$

(ii) The phase factor $\varphi(\underline{r})$ is related to the supercurrent that flows through the material below T_c

(iii) $\Psi \neq 0$ in the superconducting state, but $\Psi = 0$ above T_c

Free energy of a superconductor

The free energy of a superconductor in the absence of a magnetic field and spatial variations of n_s can be written as

$$F_s = F_n + \alpha|\Psi|^2 + \frac{\beta}{2}|\Psi|^4$$

α and β are parameters to be determined,and it is assumed that β is positive irrespective of T and that $\alpha = a(T-T_c)$ as in Landau theory

Assuming that $n_s \propto |\Psi|^2$ the equilibrium value of the order parameter is obtained from

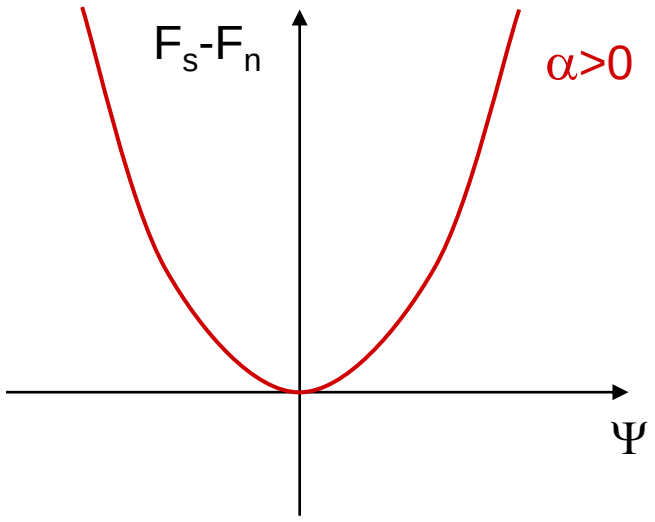
$$\frac{\partial(F_s - F_n)}{\partial n_s} = 0 = \alpha + \beta|\psi|^2$$

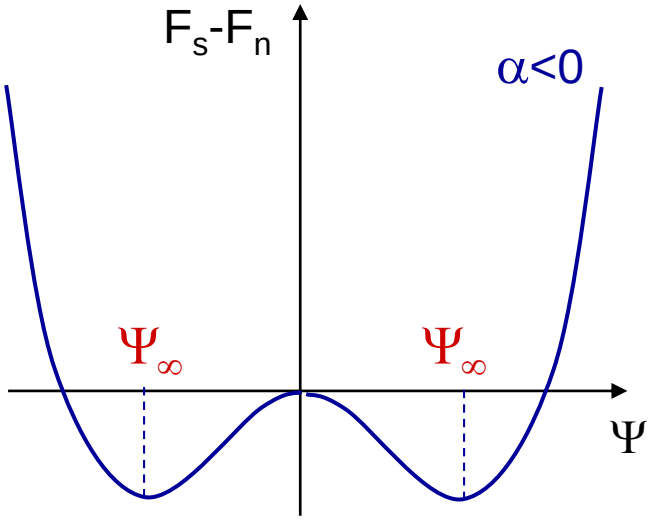
we find:

for α >0 minimum must be when $|\Psi|^2 = 0$

for α <0 minimum is when $|\Psi|^2 = \frac{-\alpha}{\beta} \equiv |\Psi_\infty|^2$

where Ψ_∞ is defined as Ψ in the interior of the sample, far from any gradients in Ψ





Free energy of a superconductor

In the superconducting state we have

$$F_s = F_n + \alpha|\Psi|^2 + \frac{\beta}{2}|\Psi|^4$$

with $$|\Psi|^2 = \frac{-\alpha}{\beta} \equiv |\Psi_\infty|^2$$

α changes sign at T_c and β is always positive for a second order transition

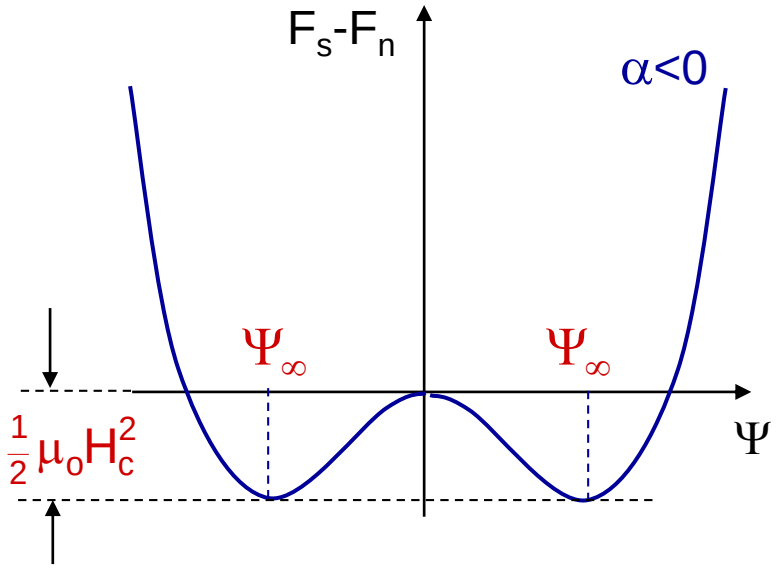
also at equilibrium

$$F_s - F_n = \frac{-\alpha^2}{2\beta}$$

But we have already shown that

$$F_s - F_n = -\frac{1}{2}\mu_o H_c^2$$

so $$\boxed{\mu_o H_c^2 = \frac{\alpha^2}{\beta}}$$



We will use this later

The full G-L free energy

If we now take the full expression for the Ginzburg-Landau free energy at a point $\underline{r}$ in the presence of magnetic fields and spatial gradients we have:

$$F_s = F_n + \alpha|\Psi|^2 + \frac{\beta}{2}|\Psi|^4$$

the term we have already discussed

$$+ \frac{1}{2}\mu_o H^2(\underline{r})$$

the magnetic energy associated with the magnetisation in a local field $H(\underline{r})$

$$+ \frac{1}{2m^*}\left|\left(-i\hbar\nabla - e^*\underline{A}\right)\Psi\right|^2$$

A kinetic energy term associated with the fact that Ψ is not uniform in space, but has a gradient

e^* and m^* are the charge and mass of the superelectrons and $\underline{A}$ is the vector potential

We should look at the origin of the *kinetic energy term* in more detail.

A charged particle in a field

Consider a particle of charge e* and mass m* moving in a field free region with velocity $\underline{v}_1$ when a magnetic field is switched on at time t=0

The field can only increase at a finite rate, and while it builds up there is an induced electric field which satisfies Maxwell's equations, ie

$$\text{curl}\,\underline{E} = -\dot{\underline{B}}$$

If $\underline{A}$ is the magnetic vector potential ($\underline{B}$=curl $\underline{A}$) then

$$\text{curl}\,\underline{E} = -\frac{d}{dt}(\text{curl}\,\underline{A})$$

Integration with respect to spatial coordinates gives

$$\underline{E} = -\frac{d\underline{A}}{dt}$$

So the momentum at time t is

$$m^*\underline{v}_2 = m^*\underline{v}_1 + \int_0^t e^*\underline{E}\,dt = m^*\underline{v}_1 - e^*\int_0^t \frac{d\underline{A}}{dt}\,dt$$

and $m^*\underline{v}_2 = m^*\underline{v}_1 - e^*\underline{A}$ or $m^*\underline{v}_1 = m^*\underline{v}_2 + e^*\underline{A}$

A charged particle in a field

If $m^* \underline{v}_2 = m^* \underline{v}_1 - e^* \underline{A}$ and $m^* \underline{v}_1 = m^* \underline{v}_2 + e^* \underline{A}$

the vector $\underline{p} = m^* \underline{v} + e^* \underline{A}$ must be conserved during the application of a magnetic field

The kinetic energy, ε, depends only upon $m^*\underline{v}$ so if $\varepsilon = f(m^*\underline{v})$ before the field is applied we must write $\varepsilon = f(\underline{p}-e^*\underline{A})$ after the field is applied

Quantum mechanically we can replace $\underline{p}$ by the momentum operator $-i\hbar\nabla$

So the final energy in the presence of a field is:

$$\frac{1}{2m^*}\left|\left(-i\hbar\nabla - e^* \underline{A}\right)\Psi\right|^2$$

Back to G-L Free Energy - 1st GL Equation

Remember that the total free energy is

$$F_s = F_n + \alpha|\Psi|^2 + \frac{\beta}{2}|\Psi|^4 + \frac{1}{2}\mu_o H^2(\underline{r}) + \frac{1}{2m^*}\left|\left(-i\hbar\nabla - e^*\underline{A}\right)\Psi\right|^2$$

This free energy, $F_s(\Psi(\underline{r}), A(\underline{r}))$, must now be minimised with respect to the order parameter, $\Psi(\underline{r})$, and *also* with respect to the vector potential $\underline{A}(\underline{r})$

To do this we must use the Euler-Lagrange equations:

(1) $$\frac{\partial F}{\partial \Psi} - \sum_j \frac{\partial}{\partial x_j}\frac{\partial F}{\partial(\nabla_j \psi)} = 0$$

(2) $$\frac{\partial F}{\partial \underline{A}} - \sum_j \frac{\partial}{\partial x_j}\frac{\partial F}{\partial(\partial A_j/\partial x_j)} = 0$$

(1) Is easy to evaluate - we only need $\frac{\partial F}{\partial \Psi} = 0$ ie

$$\alpha\Psi + \beta|\Psi|^2\Psi + \frac{1}{2m^*}\left(i\hbar\nabla + e^*\underline{A}\right)^2\Psi = 0$$

This is the First G-L equation

The second G-L equation

Evaluation of the second derivative in (2)

$$\frac{\partial F}{\partial \underline{A}} - \sum_j \frac{\partial}{\partial x_j} \frac{\partial F}{\partial (\partial A_j / \partial x_j)} = 0$$

gives $\frac{1}{\mu_o} \text{curl curl } \underline{A}$

Remember that $B = \text{curl } \underline{A}$, and that $\text{curl } \underline{B} = \mu_o \underline{J}$ where $\underline{J}$ is the current density

Therefore (2) gives

$$\underline{J} = -\frac{e^*}{m^*}\left[\Psi^2 (i\hbar\nabla + e^* \underline{A})\right]$$

This is the Second G-L equation

This is the same quantum mechanical expression for a current of particles described by a wavefunction Ψ

Magnetic penetration within G-L Theory

Taking the second GL equation: $\underline{J} = -\frac{e^*}{m^*}\left[\Psi^2(i\hbar\nabla + e^*\underline{A})\right]$ and neglecting spatial variations of Ψ:

$$\underline{J} = -\frac{e^{*2}}{m^*}\underline{A}\left|\Psi_\infty\right|^2$$

So, if curl $\underline{B} = \mu_o\underline{J}$ and using $n_s = \left|\Psi_\infty\right|^2$

$$\mathrm{curl}(\mu_o\underline{J}) = \mathrm{curlcurl}B = -\frac{\mu_o e^{*2} n^*_s}{m^*}\mathrm{curl}\underline{A}$$

and as $\mathrm{curlcurl}B = \mathrm{grad}\,\mathrm{div}B - \nabla^2 B = -\nabla^2 B$

This gives $$-\nabla^2 B = -\frac{\mu_o e^{*2} n^*_s}{m^*}B$$

and finally

$$B - \lambda^{*2}\nabla^2 B = 0$$

with

$$\lambda^{*2} = \frac{m^*}{\mu_o e^{*2} n^*_s}$$

Compare these equations directly with the London equations

A comparison of GL and London theory

We will now pre-empt a result we shall derive later in the course and recognise that superconductivity is related to the pairing of electrons.

(This was not known at the time of Ginzburg and Landau's theory)

If electrons are paired in the superconducting state then:

$$m^* = 2m_e$$

$$e^* = 2e$$

$$n^*_s = n_s/2$$

and hence $$\lambda^{*2} = \frac{2m}{\mu_o 4e^2 n_s / 2} = \frac{m}{\mu_o e^2 n_s} = \lambda_L^2$$

The coherence length

We shall now look at how the concept of the coherence length arises in the G-L Theory

Taking the 1st G-L equation in 1d without a magnetic field, ie:

$$\alpha\Psi + \beta|\Psi|^2\Psi + \frac{1}{2m^*}(i\hbar\nabla + e^*\underline{A})^2\Psi = 0$$

becomes $$\alpha\Psi + \beta|\Psi|^2\Psi - \frac{\hbar^2}{2m^*}\frac{d^2\Psi}{dx^2} = 0$$ Eq 1

Earlier we showed that the square of the order parameter can be written

$$|\Psi|^2 = \frac{-\alpha}{\beta}$$ with $\alpha<0$

However we believe that the order parameter can vary slowly with distance, so we shall now change variables and use instead a normalised order parameter

$$\Psi = f\left|\frac{\alpha}{\beta}\right|^{1/2}$$

where f varies with distance

The coherence length

Substituting the normalised order parameter f into equation 1 on the previous slide, and noting that $|\Psi|^2 = -\alpha/\beta$, we obtain a "non linear Schrodinger equation"

$$-\frac{\hbar^2}{2m^*}\left|\frac{\alpha}{\beta}\right|^{1/2}\frac{d^2f}{dx^2}+\alpha f\left|\frac{\alpha}{\beta}\right|^{1/2}-\beta f^3\frac{\alpha}{\beta}\left|\frac{\alpha}{\beta}\right|^{1/2}=0$$

"-" disappears as $\alpha<0$ *and we introduce* $|\alpha|$

Canceling gives $$-\frac{\hbar^2}{2m^*}\frac{d^2f}{dx^2}+\alpha f-\beta f^3\frac{\alpha}{\beta}=0$$ hence $$\frac{\hbar^2}{2m^*|\alpha|}\frac{d^2f}{dx^2}+f-f^3=0$$

Making the substitutions $f=1+f'$ where f' is small and negative, and $\xi^2=\frac{\hbar^2}{2m^*|\alpha|}$

we have $$\xi^2\frac{d^2f'}{dx^2}+1+f'-(1+3f'.....)=0$$

hence $$\boxed{\frac{d^2f'}{dx^2}=\frac{2f'}{\xi^2}}$$

the solution of which is $f'(x)=\exp(-x\sqrt{2}/\xi)$

ξ is therefore the coherence length, characteristic distance over which the order parameter Ψ varies

Relationship between B_c, λ^* and ξ

To summarise we have

$$|\alpha| = \frac{\hbar^2}{2m^*\xi^2} \quad (1) \qquad |\Psi|^2 = n_s = \frac{m^*}{\mu_o e^{*2}\lambda^{*2}} = -\frac{\alpha}{\beta} \quad (2) \text{ and } \quad \mu_o H_c^2 = \frac{\alpha^2}{\beta} \quad (3)$$

Solving for α and β

$$\alpha = -\frac{\mu_o^2 H_c^2 e^{*2}\lambda^{*2}}{m^*} \quad \text{from (2) and (3)}$$

$$\beta = -\frac{\mu_o^3 H_c^2 e^{*4}\lambda^{*4}}{m^{*2}} \quad \text{by substituting in (2)}$$

Finally, using (1) we have

$$(\mu_o H_c)^2 \xi^2 \lambda^{*2} = \frac{1}{2}\left(\frac{\hbar}{e^*}\right)^2 = \text{constant}$$

or

$$B_c^2 \xi^2 \lambda^{*2} = \text{constant}$$

So, although B_c, λ^* and ξ are all temperature dependent, their product is not

although experimentally it is found to be not quite independent of T